SHINING 3D

PRECISE3DM

SHINING 3D
METROLOGY

SHINING 3D Metrology Solutions

Going Beyond Precision

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High-Accuracy 3D Vision Technology



Accuracy Laboratory

The SHINING 3D Accuracy Laboratory is accredited to ISO/IEC 17025 by CNAS. The laboratory is equipped with a comprehensive suite of calibration instruments and conducts calibration and verification services in strict accordance with internationally recognized standards, including VDI/VDE 2634 and ISO 10360. Through these capabilities, the laboratory provides authoritative technical support and reliable quality assurance to partners and customers worldwide.









ISO 17025 CERTIFIED

ilac-MRA CNAS

FreeScan Omni Series

World's First Standalone Inspection-Ready Metrology 3D Scanner

worldskills



Metrology-Grade On-Scanner Inspection

From scanning and meshing to inspection and reporting, every step is seamlessly completed right on the scanner. With preset templates and an intuitive interface, frontline staff can just scan the part and get instant, consistent reports—making quality control efficient and scalable.



Flexible Expansion for Evolving Needs

FreeScan Omni completes scanning, inspection, and reporting in one flagship configuration. FreeScan Omni Lite shares the same hardware foundation as FreeScan Omni. As application needs evolve, FreeScan Omni Lite can be expanded with dedicated modules.



Standalone & Wireless Design

With a built-in touchscreen, FreeScan Omni offers a truly wireless experience—no cords, no data cables. The full workflow runs directly on the device.



AI feature recognition

Intelligent boundary detection provides fast, precise measurement of round holes, rectangular holes, and slots, ensuring high-quality and high-accuracy data capture.

Technical Specification

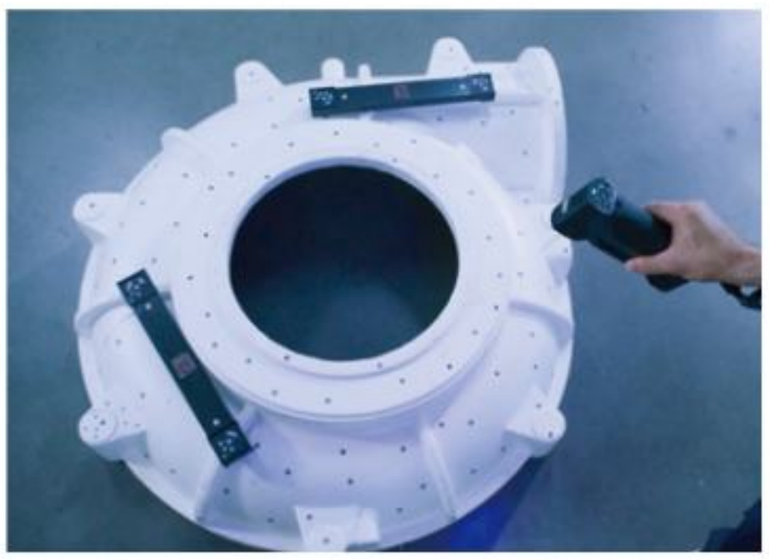
	FreeScan Omni Series
High-speed scan	Included (93 laser lines) | VCSEL
Detailed scan	Included (25 parallel laser lines)
Deep pockets scan	Included (1 laser line)
Accuracy	0.02 mm
Volumetric accuracy with VPG	0.02 + 0.015 mm/m
Max. FOV	Laser: 580 x 650 mm; IR: 1205 x 1104 mm
Scan speed	Up to 7,619,000 points/s
Resolution	0.01 ~ 10 mm
Depth of field	830 mm (170 ~ 1000 mm)
Net weight	≤1.1 kg
Certifications	CE, FCC, ROHS, WEEE, KC, FDA, UKCA, IP50,TELEC, TiSAX
Acceptance test	VDI/VDE 2634 Part 3, ISO 10360 (Tested in the ISO/IEC 17025 Accredited Accuracy Lab)

FreeScan Combo Series

FreeScan Combo+/Combo Wireless

Wireless Hybrid Light Source Metrology
Handheld 3D Scanner





Wireless with Reliable Data Transfer

Powered by next-generation Wi-Fi 7 and advanced anti-interference technology, FreeScan Combo+/Combo Wireless delivers fast, stable wireless data transmission across factory floors, confined workspaces, and outdoor environments.

Built-in Video Photogrammetry (VPG)

Integrated patented Video Photogrammetry (VPG) ontinuously optimizes the spatial position of reference markers during scanning, minimizing cumulative errors and maintaining reliable volumetric accuracy—even for large components.





Compact and Powerful

Weighing only 550 g, FreeScan Combo+/Combo Wireless series boasts an ergonomic, compact, and lightweight design that makes 3D scanning convenient and easy.
The optimized camera angle makes it better suited to working in narrow, complex environments, enabling fast data capture along seams.

Marker-Free Infrared Scanning

The infrared scanning mode minimizes surface preparation by reducing the need for markers. Ideal for medium-sized parts, it streamlines the scanning workflow while maintaining reliable geometry capture for reverse engineering and industrial applications.

Technical Specification

	FreeScan Combo+/Combo Wireless
High-speed scan	FreeScan Combo+ Wireless: 93 laser lines | VCSEL; FreeScan Combo Wireless: 50 laser lines | VCSEL
Detailed scan	25 parallel laser lines
Deep pockets scan	1 laser line
Light source	Blue Laser & Infrared VCSEL
Accuracy	0.02 mm
Max. FOV	Laser: 580 x 650 mm; IR: 830 x 700 mm
Scan speed	FreeScan Combo+ Wireless: Up to 9,106,000 points/s; FreeScan Combo Wireless: Up to 7,344,000 points/s
Volumetric accuracy with VPG	0.02 + 0.015 mm/m
Resolution	0.01 ~ 10 mm
Net weight	550 g
Certifications	CE, FCC, ROHS, WEEE, KC, FDA, UKCA, IP50, TELEC, TISAX
Acceptance test	ISO 10360 (certified in ISO 17025 certificated accuracy lab)

FreeScan Combo Series

FreeScan Combo+/Combo

Hybrid Light Source and Multifunctional Handheld 3D Scanner

worldskills

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Blue Laser + Infrared Light

An innovative combination of blue laser and infrared light sources meet the needs of diverse industrial 3D scanning scenarios.



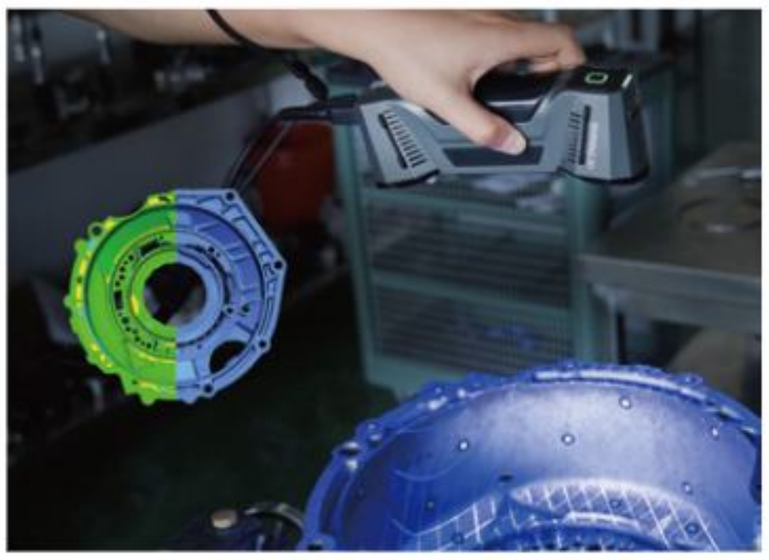
Infrared Mode Scan without Markers

No markers needed for feature-rich workpieces. Infrared light source can be used directly to scan surfaces quickly.



Working in Confined Spaces

FreeScan Combo+/Combo is able to work in confined spaces. The lens angle is optimized to acquire full data on narrow seams.



Light Weight and Compact

Small size of 193 x 63 x 53 mm and weight only of 620g.



Technical Specification

	FreeScan Combo+/Combo
High-speed scan	FreeScan Combo+: 50 laser lines | VCSEL; FreeScan Combo: 26 laser lines | VCSEL
Detailed scan	7 parallel laser lines
Deep pockets scan	1 laser line
Light source	Blue Laser & Infrared VCSEL
Accuracy	0.02 mm
Max. FOV	520 x 510 mm / 600 x 600 mm
Volumetric accuracy	0.02 + 0.033 mm/m
Point distance	0.05 ~ 10 mm / 0.1 ~ 3 mm
Scan speed	Up to 3,600,000 points/s
Dimension	193 x 63 x 53 mm
Certifications	CE, FCC, ROHS, WEEE, FDA, UKCA, IP50, KC, TiSAX
Accuracy certification	VDI/VDE 2634 Part 3, ISO 10360 (Tested in the ISO/IEC 17025 Accredited Accuracy Lab)

FreeScan UE Pro Series

Multi-Functional Laser Handheld 3D Scanner

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FreeScan UE Pro2

50 Laser Lines

FreeScan UE Pro

26 Laser Lines

Technical Specification

	FreeScan UE Pro Series
Light source	Blue Laser
Accuracy	0.02 mm
Max. FOV	600 × 550 mm
Volumetric accuracy	0.02 + 0.03 mm/m (standard mode) - 0.02 + 0.015 mm/m (built-in photogrammetry mode)
Point distance	0.01 ~ 10 mm
Scan speed	Up to 3,460,000 points/s
Dimension	298 x 103.5 x 74.5 mm (FreeScan UE Pro2) / 305.8 x 118.9 x 100.8 mm (FreeScan UE Pro)
Weight	950 g (FreeScan UE Pro2) / 840 g (FreeScan UE Pro)
Certifications	CE, FCC, ROHS, WEEE, KC, FDA, UKCA, IP50, TELEC, TiSAX
Accuracy certification	VDI/VDE 2634 Part 3, ISO 10360 (Tested in the ISO/IEC 17025 Accredited Accuracy Lab)



FreeScan UE Pro2

Inspect on the move

Wireless and Portable

Integrated with Wireless module, FreeScan UE Pro2 liberates you from cables, allowing seamless data transmission during scanning.

Double the Speed, Double the Productivity

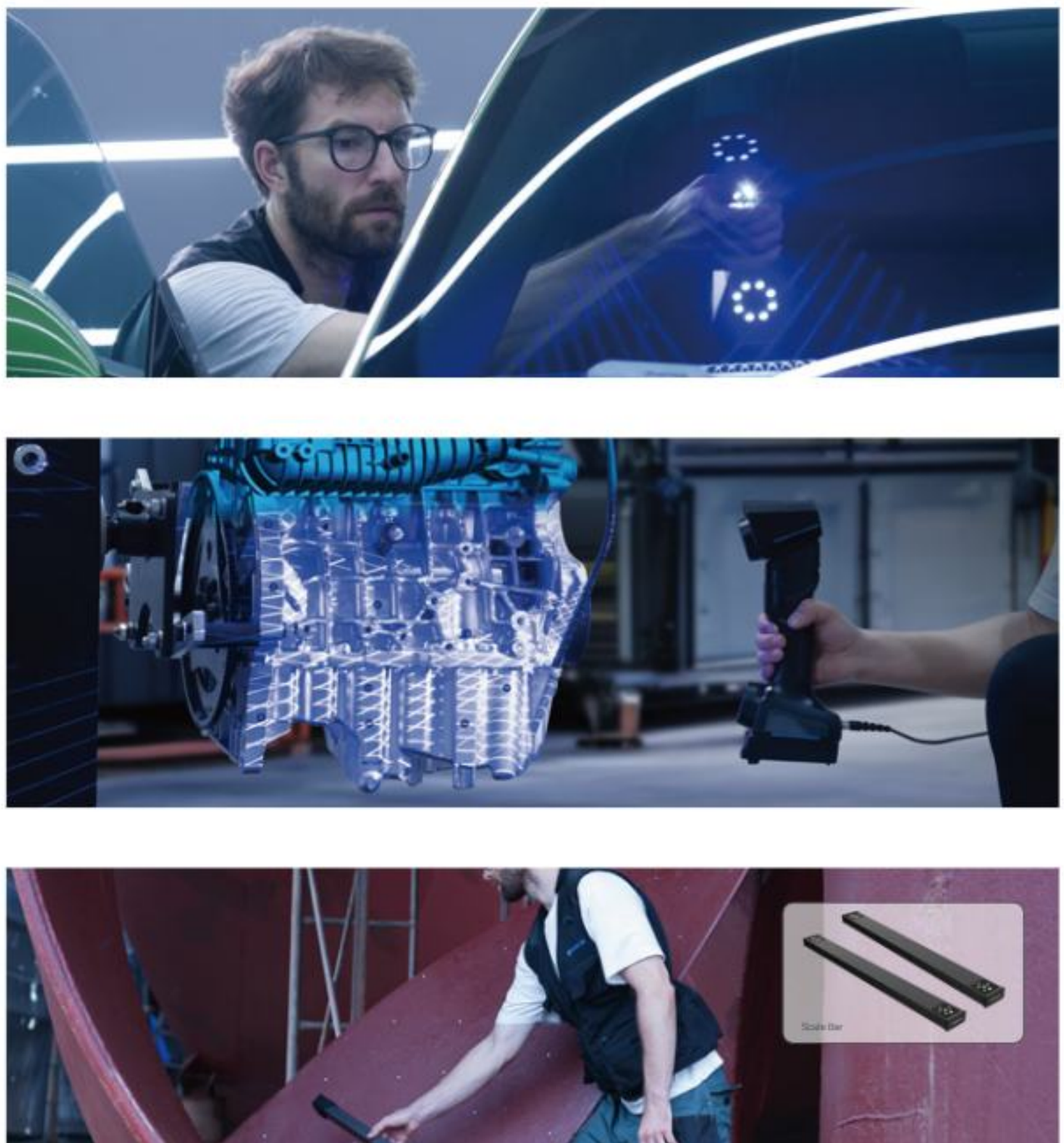
FreeScan UE Pro2 offers nearly double scanning efficiency compared to the previous generation. With a lightning-fast scan speed up to 3,460,000 points/s.

Certified and Guaranteed Results

0.02 mm accuracy delivers consistent scanning results with high precision.

Binocular Photogrammetry: Elevated Accuracy for Large-scale Scans

FreeScan UE Pro series feature patented innovative built-in photogrammetry technology for scanning large objects, quickly optimizing the spatial position of global markers, ensuring consistent accuracy.



FreeScan UE Nova Series

Wireless Metrology 3D scanner for greater reach



Wide-Range Laser Scan: Vast and Fast

FreeScan UE Nova series offers expansive coverage, delivering unparalleled efficiency and flexibility.



Flexible FOV

FreeScan UE Nova series offers the flexible field of view depending on three selectable working range, adapt to different on-site operation conditions.

- Near Mode (300 ~ 800 mm)
- Standard Mode (600 ~ 1500 mm)
- Far Mode (1200 ~2600 mm)



Expand Possibilities with Trak Nova Serie

Elevate your capabilities and tackle even the most complex work environments with FreeScan Trak Nova series.

- Dynamic tracking
- Marker-free scanning



Video Photogrammetry (VPG)

Utilizing SHINING 3D's patented video-based photogrammetry (VPG) without coded markers, it integrates with a calibration rod for real-time marker verification, delivering volumetric accuracy of 0.072 mm + 0.012 mm/m for more efficient and reliable data acquisition.

FreeScan UE Nova+

62 Laser Lines

5,714,000 points/s

FreeScan UE Nova

50 Laser Lines

4,600,000 points/s

Technical Specification

	FreeScan UE Nova Series
Light source	Blue Laser
Accuracy	0.072 mm
Max. FOV	2.6 × 2.2 m
Volumetric accuracy with VPG	0.072 + 0.012 mm/m
Point distance	0.5 ~ 10 mm
Depth of field	2300 mm (300 ~ 2600 mm)
Net Weight	1.6 kg
Certifications	CE, FCC, ROHS, WEEE, KC, FDA, UKCA, IP50, TELEC, TISAX
Acceptance test	VDI/VDE 2634, ISO 10360 (certificated in ISO 17025 certificated accuracy lab)

FreeScan Trio

First Marker-Free Laser 3D Scanner



98 Laser Lines, Zero Markers

The FreeScan Trio's thorough 98-laser-line mode eliminates the need for markers, efficiently capturing all your workpieces.



Three 5 Megapixel Cameras

With 3 industrial cameras, each wielding 5 megapixels, FreeScan Trio brings out the most elaborate details and high-quality 3D data.



Unrelenting Accuracy and Precision

For quality control, inspection, and reverse engineering, its 0.02 mm accuracy (with markers) and high-precision performance deliver results you can count on, time and time again.



Patented Photogrammetry

The FreeScan Trio features a binocular photogrammetry mode that can achieve a remarkable volumetric accuracy of up to 0.02 mm + 0.015 mm/m.

Technical Specification

	FreeScan Trio
Light source	26 laser lines, Single laser line, 7 parallel laser lines, 98 laser lines
Accuracy	0.02 mm
Max. FOV	650 x 580 mm
Volumetric accuracy	0.02 + 0.03 mm/m (0.02 + 0.015 mm/m with photogrammetry)
Point distance	0.01 ~ 3 mm
Scan speed	Up to 3,010,000 points/s
Dimension	331 x 120 x 76 mm
Weight	985 g
Certifications	CE, FCC, ROHS, WEEE, KC, FDA, UKCA, IP50, TiSAX
Accuracy certification	VDI/VDE 2634 Part 3, ISO 10360 (Tested in the ISO/IEC 17025 Accredited Accuracy Lab)

FreeScan Trak ProW+

Wireless 3D Dynamic Tracking and Scanning System
with Extended Range

Technical Specification

	FreeScan Trak ProW+
High-speed scan	FreeScan Trak ProW+: 93 laser lines; FreeScan Trak ProW: 50 laser lines
Detailed scan	FreeScan Trak ProW+: 25 laser lines; FreeScan Trak ProW: 7 laser lines
Deep pockets scan	1 laser line
Accuracy	0.023 mm
Volumetric accuracy with VPG	0.044 + 0.012 mm/m
Scan speed	FreeScan Trak ProW+: 7,600,000 points/s; FreeScan Trak ProW: 5,500,000 points/s
Resolution	0.01 ~ 10 mm
Connection	Wireless & Wired mode (fiber optic)
Certifications	CE, FCC, ROHS, WEEE, KC, FDA, UKCA, IP50, TELEC, TISAX
Acceptance test	VDI/VDE 2634 Part 3, ISO 10360 (certificated in ISO 17025 certificated accuracy lab)









Ultra-Wide Measuring Range

FreeScan Trak ProW+ is designed for efficient single-station measurement of large workpieces.FreeScan Trak ProW+ features a groundbreaking design with an ultra-long 8.6 tracking distance and widely expands the single-station tracking volume to 156 m³, significantly reducing the need for leapfrog during large-scale object scanning.

Wireless Scanning Wherever You Go

FreeScan Trak ProW+ delivers fast, wireless scanning anywhere. Its cable-free design enables real-time data transfer, with power bank support for unrestricted mobility. An integrated screen ensures seamless operation.

No Markers Needed, Fast Paced Scanning

The tracker enables real-time scanner recognition and positioning, eliminating the need for reference markers, which enhances scanning efficiency and streamlines the workflow.

Video Photogrammetry (VPG)

FreeScan Trak ProW+ features SHINING 3D's patented video-based photogrammetry, which eliminates the need for coded markers. This ensures consistent volumetric accuracy and streamlines the setup process for efficient large-object scanning.

FreeScan Trak Nova Series

Wireless Multi-Functional
Dynamic Tracking and Scanning System

DESIGN AWARD 2026

TE Nova

UE Nova

Technical Specification

	FreeScan Trak Nova Series
High-speed scan	FreeScan Trak Nova+: 62 laser lines; FreeScan Trak Nova: 50 laser lines
Detailed scan	FreeScan Trak Nova+: 25 laser lines; FreeScan Trak Nova: 7 laser lines
Deep pockets scan	1 laser line
Accuracy	0.02 mm
Volumetric extension accuracy with VPG	0.046 + 0.012 mm/m
Resolution	0.01 ~ 10 mm
Scan speed	7,600,000 points/s
Flexible FOV	Up to 2600 × 2200 mm
Connection	Wireless & Wired mode (fiber optic)
Certifications	CE, FCC, ROHS, WEEE, KC, FDA, UKCA, IP50, TELEC, TISAX
Acceptance test	VDI/VDE 2634 Part 3, ISO 10360 (certificated in ISO 17025 certificated accuracy lab)



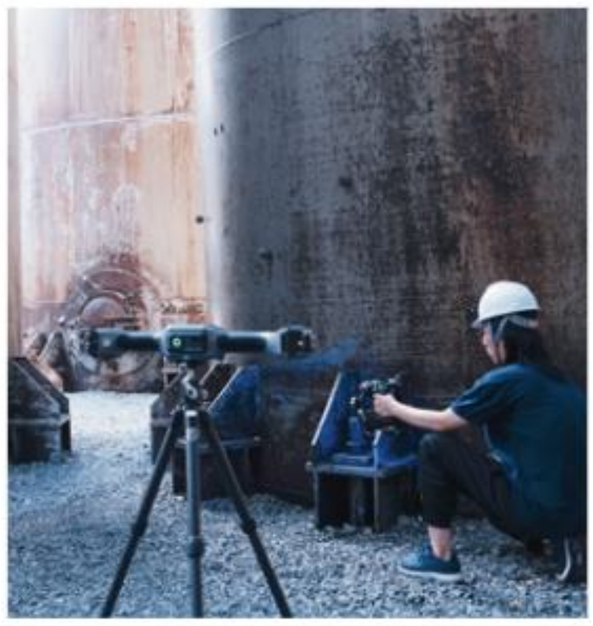
Dynamic Tracking Scan: Compact and Agile

Real-time dynamic tracking enables fast, flexible, and marker-free 3D scanning. Combined with its wireless, compact design, the system excels at capturing fine details in confined spaces while delivering efficient and precise measurements.



Flexible & Separable System

FreeScan Trak Nova series includes a detachable handheld scanner. The tracker, FreeScan UE Nova, also serves as a handheld laser scanner with the largest FOV on the market, delivering unparalleled efficiency and flexibility.



Engineered to Boost Productivity

FreeScan Trak Nova+ achieves scanning speeds of up to 7,600,000 points/s, enabling rapid acquisition of high-quality 3D data while maintaining fine detail.



Keep Going with Swappable Batteries

For maximum convenience, FreeScan Trak Nova Series features detachable batteries for extended scanning time. Easily swap out batteries to keep your workflow uninterrupted, even during long projects or in remote locations.

FreeProbe Series

Portable Probe Solution for Shop Floor Measurement



Wireless and Portable

Move freely without cables that enhance usability in complex environments and improves overall workflow efficiency.



Multiple Tip Options

Two probe tips of different diameters to suit a variety of measurement tasks. Each tip can be further extended or adapted another type for greater reach.



Deeper Probing Capability

FreeProbe 2 Pro can maintain stable probing performance even when some marker plates are blocked, ensuring uninterrupted scanning and inspection.



Ergonimic Design

Designed with a lightweight body and ergonomic grip, reducing fatigue during prolonged use.

Technical Specification

	FreeProbe 2		FreeProbe 2 Pro
Compatible trackers	FreeScan Trak Nova Series	FreeScan Trak ProW Series	FreeScan Trak ProW Series
Accuracy	0.025 mm		
Volumetric accuracy for extension	0.046 + 0.012 mm/m		0.044 + 0.012 mm/m
Weight	460 g		590 g
Dimensions	50 x 110 x 363 mm		50 x 113.5 x 454.5 mm
Accuracy certification	ISO 10360 (Tested in the ISO/IEC 17025 Accredited Accuracy Lab)		

OptimScan Q Series

OptimScan Q12/Q9 HD

High-Precision 3D Inspection Scanner



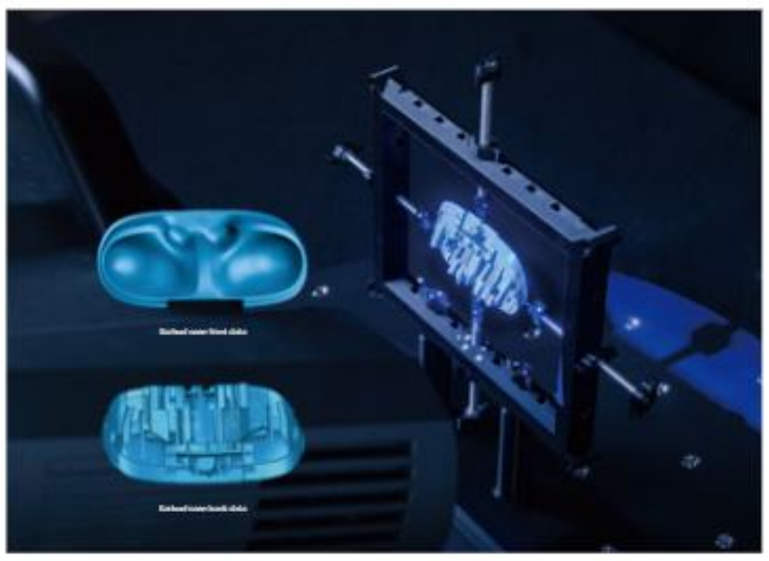
Accuracy with Certified Metrology Confidence

OptimScan Q12/Q9 HD pushes beyond the performance limits of fringe projection optical scanning systems, delivering a new level of accuracy and fine-detail capture within its class.



Effortless Scan Range Switching

OptimScan Q12/Q9 HD features two scan ranges that can be switched with a single click, eliminating manual adjustment of aperture and focal length. Scanned data from different ranges can be seamlessly fused in the software, ensuring both efficiency and detail.



Monocular-Stereo Fusion (MSF)

The monocular-stereo fusion function can effectively increase data coverage and improve the integrity of the scanned data, such as the common turning corners and joints of industrial parts.



Automated Inspection Solution

OptimScan Q12/Q9 HD can seamlessly integrate with intelligent 3D inspection system, enabling fully automated workflow from scanning to report generation.

Workflow

Path Teaching → Automated 3D Measurement → Automated 3D Inspection → Automated Report Generation



Technical Specification

	OptimScan Q12/Q9 HD	
Scan range	Large range	Small range
FOV	220 × 150 mm	80 × 50 mm
Accuracy	0.01 mm	0.004 mm
Point distance	OptimScan Q12 HD: 0.054 mm; OptimScan Q9 HD: 0.06 mm	OptimScan Q12 HD: 0.02 mm; OptimScan Q9 HD: 0.03 mm
Working distance	600 mm	200 mm
Depth of field	120 mm	30 mm
Resolution	OptimScan Q12: 4 × 12.3MP; OptimScan Q9: 4 × 9MP	
Light source	Blue LED	
Weight	3.6 kg	
Dimensions	366 × 162 × 132 mm	
Certifications	CE, FCC, ROHS, WEEE, KC, FDA, UKCA, IP50, TELEC, TISAX	
Acceptance test	VDI/VDE 2634 Part 2, ISO 10360 (certified in ISO 17025 certificated accuracy lab)	

OptimScan Q Series

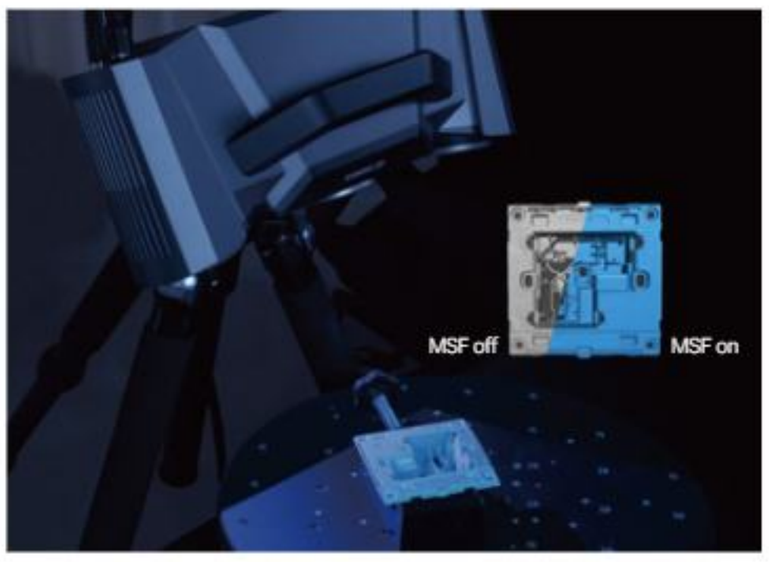
OptimScan Q12/Q9

High-Precision 3D Inspection Scanner



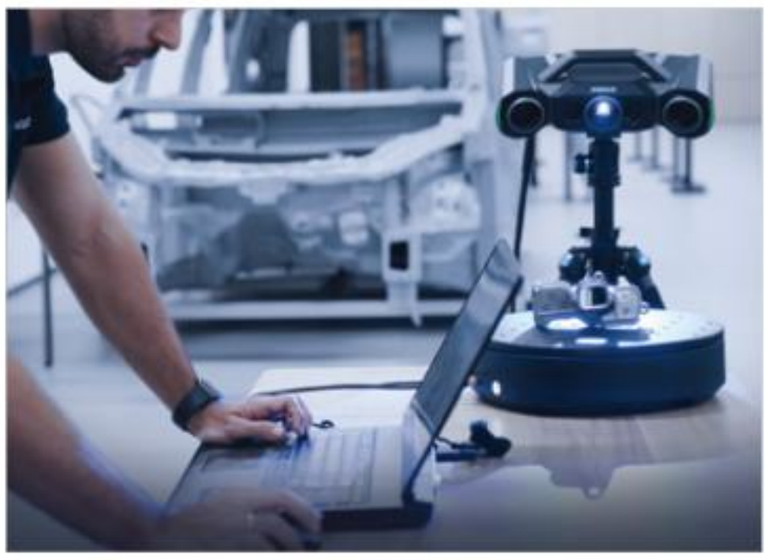
Reliable High Performance

Equipped with four high-resolution cameras, OptimScan Q12/Q9 can accurately capture geometrically intricate features on an object's surface. Combined with powerful hardware modules and advanced 3D reconstruction algorithms, the systems deliver an impressive accuracy of 0.005 mm, which has been verified in accordance with ISO 10360-13, ensuring stable and reliable repeatability across a wide range of applications.



Monocular-Stereo Fusion (MSF)

The monocular-stereo fusion function can effectively increase data coverage and improve the integrity of the scanned data, such as the common turning corners and joints of industrial parts.



Multiple Exposure, Faster Scanning

Powered by a built-in dual-chip computing module, OptimScan Q12/Q9 delivers rapid image acquisition and efficient data processing. Each shot takes less than 1 second. To further streamline the workflow, the systems support multiple exposure modes, significantly reducing scanning time while ensuring optimal results across a wide range of surface conditions.



Automated Inspection Solution

OptimScan Q12/Q9 can seamlessly integrates with intelligent 3D inspection system, enabling fully automated workflow from scanning to report generation.

Workflow

Path Teaching → Automated 3D Measurement → Automated 3D Inspection → Automated Report Generation

Technical Specification

	OptimScan Q12/Q9	
Scan range	Large range	Small range
FOV	430 × 300 mm	160 × 110 mm
Accuracy	0.015 mm	0.005 mm
Point distance	OptimScan Q12: 0.1 mm, OptimScan Q9: 0.11 mm	OptimScan Q12: 0.04 mm, OptimScan Q9: 0.05 mm
Working distance	590 mm	210 mm
Depth of field	300 mm	60 mm
Resolution	OptimScan Q12: 4 × 12.3MP, OptimScan Q9: 4 × 9MP	
Light source	Blue LED	
Weight	3.5 kg	
Dimensions	366 × 162 × 132 mm	
Certifications	CE, FCC, ROHS, WEEE, KC, FDA, UKCA, IP50, TELEC, TiSAX	
Acceptance test	VDI/VDE 2634 Part 2, ISO 10360 (Tested in the ISO/IEC 17025 Accredited Accuracy Lab)	

AutoScan Inspec2

Automatic Desktop 3D Scanner

Automatic and Efficient

Features 3-axis turntable, enabling hands-free, multi-angle scanning with one-click operation to capture data efficiently.

Metrology-Grade Accuracy

With a scan accuracy of 0.01 mm, it ensures consistent, reliable results for complex small objects.

Unparalleled Details

Equipped with 2 × 5MP high resolution cameras, it can capture every detail of the components meticulously.





Technical Specification

	AutoScan Inspec2
Accuracy	0.01 mm
Field of view	140 × 90 × 80 mm
Align mode	Feature, Markers
Point distance	0.05 mm
Rotation axis	3 Axis
Camera resolution	2 × 5MP
Light source	LED blue light
Color scan	Supported
Dimensions	420 × 260 × 270 mm
Weight	5 kg
Certification	CE, FCC, ROHS, WEEE, KC

Automation Solution

RobotScan Series

RobotScan series is engineered for replacing manual operations with robotic automation. It enables fully automated and standardized 3D scanning to rapidly acquire precise geometric data and automatically generate detailed inspection reports. The system supports customized configurations for specific inspection needs and integrates with MES systems for real-time feedback and secure data management, enhancing the efficiency and quality of intelligent manufacturing.

Standard Automation Solutions



RobotScan Q12

- Fringe projection technology with the best accuracy and high detailed data
- Ideal for small-sized part measurement with complex surfaces, geometries, or sharp edges
- Fine -detail components such as airfoils, brackets, connectors, gears, housings, and electronic components
- Automated optical 3D measurement solution with high repeatability and high efficiency for mass-production inspection



RobotScan UE Pro2

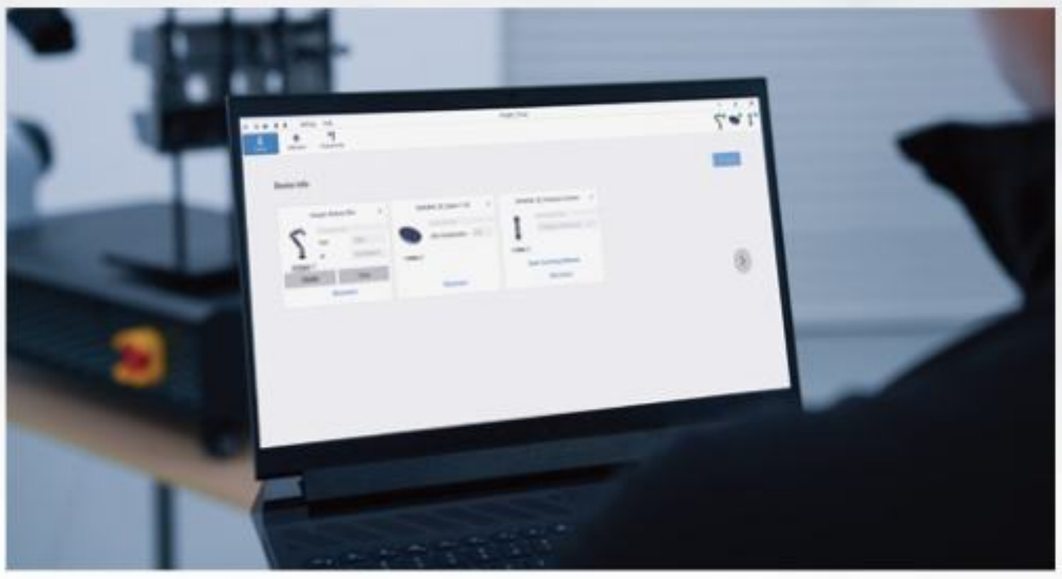
- Handheld blue laser scanning technology with strong material adaptability to black, shiny, and reflective surfaces
- Suited for small and medium sized sheet metal parts, casting housing, machining parts, mold components
- Wireless and Wired connectivity for flexible on-site integration
- Automation optical measurement solution with excellent balance of accuracy and efficiency



RobotScan Combo+

- Hybrid light source delivers excellent versatility across different part types.
- Efficient batch inspection of the small-to-medium-sized castings and machined parts
- Mixed-size components such as covers, trim pieces, assemblies, medium castings
- Cost-effective automated 3D scanning solution for manufacturers with diverse inspection needs.

Control & Inspection Software



RobotScan Control

RobotScan Control is a professional automation software purpose-built for industrial applications. It enables direct control of collaborative robot arms, servo turntables, and 3D scanners to execute the complete workflow - from path programming to 3D scanning and inspection. Featuring a process-oriented interface, the software offers intuitive operation and a low learning curve, greatly improving automation efficiency and stability.



SHINING3D Inspect

SHINING3D Inspect is an industrial inspection software certified by PTB for measurement accuracy. It processes high-accuracy 3D scan data to perform detailed 3D comparison, measurement, and inspection report generation. The software delivers reliable, traceable inspection results that meet the stringent demands of modern manufacturing and quality control.

Customized Solutions

RobotScan series also offers a fully customizable configuration to meet the diverse 3D measurement needs of multiple industries. Components such as different brands of robotic arm, SHINING 3D scanner, turntable, fixtures, and control and inspection software can all be tailored to specific application requirements.



RobotScan Trak nova



RobotScan with AGV



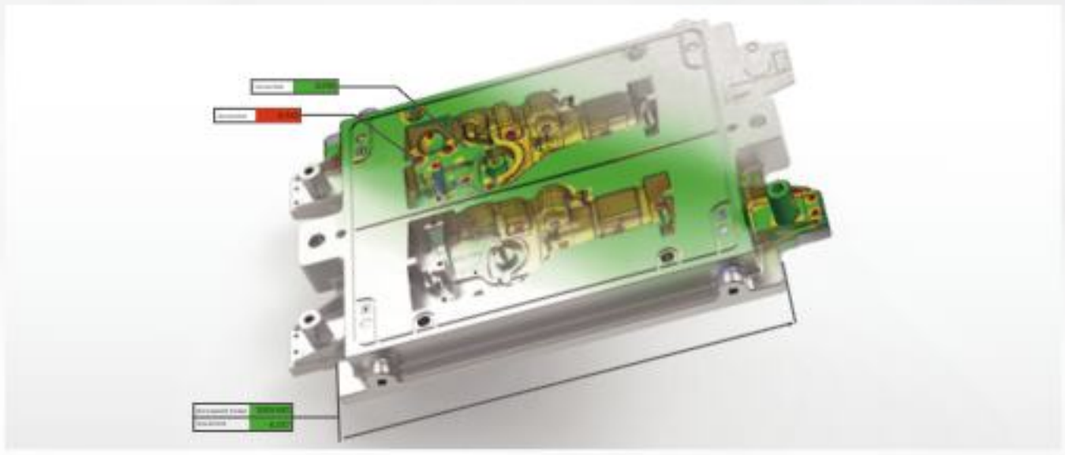
Automated Loading & Unloading System



RobotScan with Servo Gantry

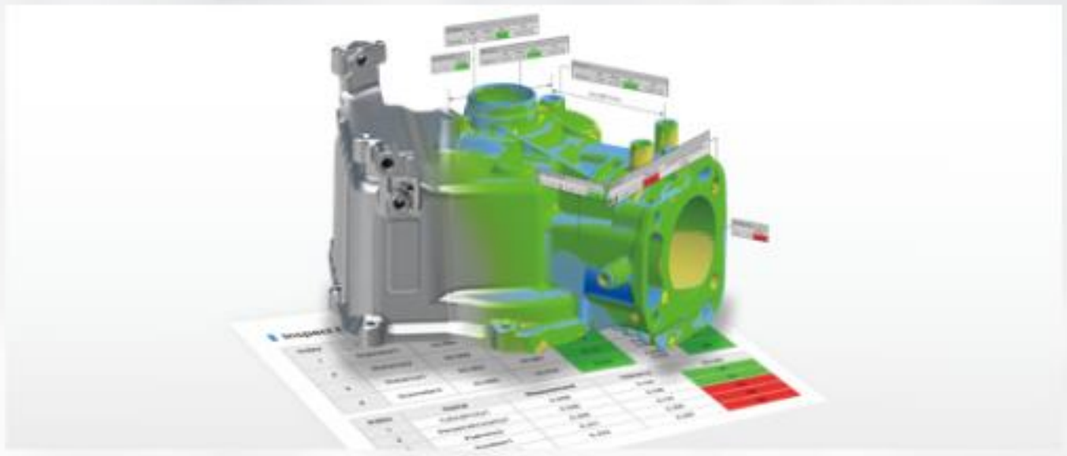
3D Inspection Software

SHINING3D Inspect



Full-Field Deviation Analysis

- Perform full-field comparison between scan data and CAD
- Visualize deviations using 3D color maps
- Analyze cross-sectional deviations in 2D
- Customize color scales for clearer and more flexible interpretation



Standard-Compliant GD&T Evaluation

- Switch between ISO and ASME standards
- Evaluate critical dimensions based on GD&T standards
- Support tolerance verification for functional features



Application-Oriented Inspection Modules

Dent Inspection
Identify dents and surface deviations efficiently.

Sheet Metal Inspection
Analyze deformation, springback, and profile deviations.



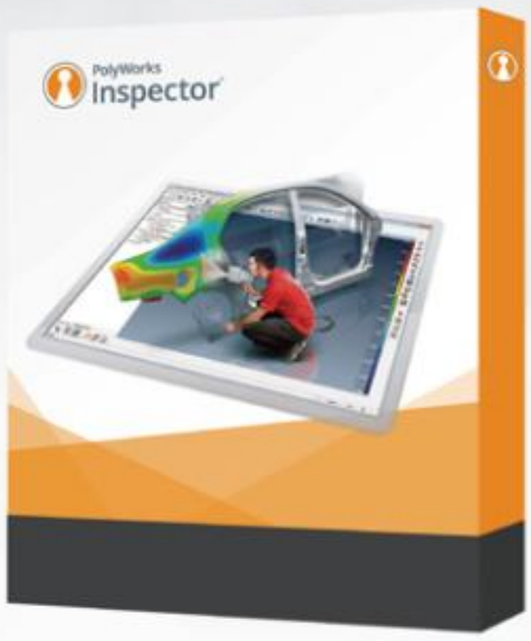
Automated and Customizable Reporting

- Turn inspection results into clear, structured deliverables
- Visualize deviations and measurement results intuitively
- Support quality documentation with full traceability

Key Features

- Multi-Format Data Import and Management
- Multi Alignment Toolset
- 2D Dimensions Inspection
- 2D & 3D Comparison
- Measurement Toolset
- Intelligent Feature Recognition
- Cross-Section Analysis Toolset
- Light/Dark UI Available

3D Inspection Software



PolyWorks | Inspector

The 3D dimensional analysis and quality control solution to gain control of your product engineering and manufacturing process.

Powerful Measurement and Analysis Capabilities
Comprehensive Compatibility and Integration
Automated and Customizable Workflows

3D Design Software



EXModel

EXModel is a powerful bridge that simplifies CAD modeling from 3D scanning to manufacturing. It provides a comprehensive set of tools that enable you to transform mesh into a professional quality CAD solid model in easy steps.

Effortless Mesh Processing
Precision CAD Creation
Enhanced Data Utilization

Texture Mapping Software



BlueStar Mapping

Through intelligent and semi-automated workflows, BlueStar Mapping precisely maps multi-angle photos captured by smartphones or DSLR cameras onto 3D scanned models, generating 3D models with high precision, exceptional clarity, and true-color textures.

Universal Compatibility
One-Stop Workflow
High Definition Resolution

Metrology Accessories

Markers Kit Series

Mark Your Scan, Master Your Accuracy



Simplify Scanning

Magnetic marker balls for fast setup and easy positioning, Metrology Kit is perfect for handheld laser 3D scanners with smooth scanning workflows.



Accuracy in Every Angle

Markers Kit for Dynamic Tracking System supports stable and precise tracking with different leapfrog marker options for scanning large and complex objects.

Specification

Markers Kit for Handheld Scanners	Markers Kit for Dynamic Tracking System
15 × Magnetic marker balls (6mm)	12 × Magnetic marker balls (12mm)
10 × Magnetic markers (6mm, 50 markers / sheet)	14 × Magnetic markers (12mm, 15 markers / sheet)
20 × Markers (6mm, 50 markers / sheet)	50 × Markers (12mm, 15 markers / sheet)
20 × Marker cards	12 × 45-degree leapfrog markers (12mm)
1 × Carrying case	12 × 90-degree leapfrog markers (12mm)
	1 × Carrying case

Automatic Dual-Axis Turntable

Automated & Simplified Operation

The dual-axis turntable performs automatic rotation and tilting motions, enabling more complete data capture in a single scan. This enhances data acquisition efficiency and supports batch 3D inspection workflows.

Complete Data Acquisition

The dual-axis turntable captures 3D data from multiple angles. With the intelligent add-scan function, it can automatically plan and fill in missing areas.

Versatile Workflow

Supports flip scanning with automatic alignment for full coverage. Also enables simultaneous multi-object scanning with flexible region segmentation and per-object export, boosting overall efficiency.

Flexible and Portable

Featuring an all-in-one design, the turntable is easy to transport and use. It can be paired with high-precision 3D inspection systems for on-site inspections in production workshops.

Specification

	Automatic Dual-Axis Turntable
Turntable type	Two-axis (Azimuth axis + Tilt axis)
Dimensions	340 × 238 × 273 mm
Motion range	Azimuth axis: 360° rotation; Tilt axis: ±90° rotation
Maximum workpiece size	200 mm
Maximum load	3 kg
Weight	3.6 kg
Interface	USB 2.0
Power	Max. 30 W
Voltage range	90 ~ 240 VAC, 50 ~ 60 Hz
Working temperature & humidity	-10 ~ 40°C; 10% ~ 90% RH
Control unit	Start / Pause



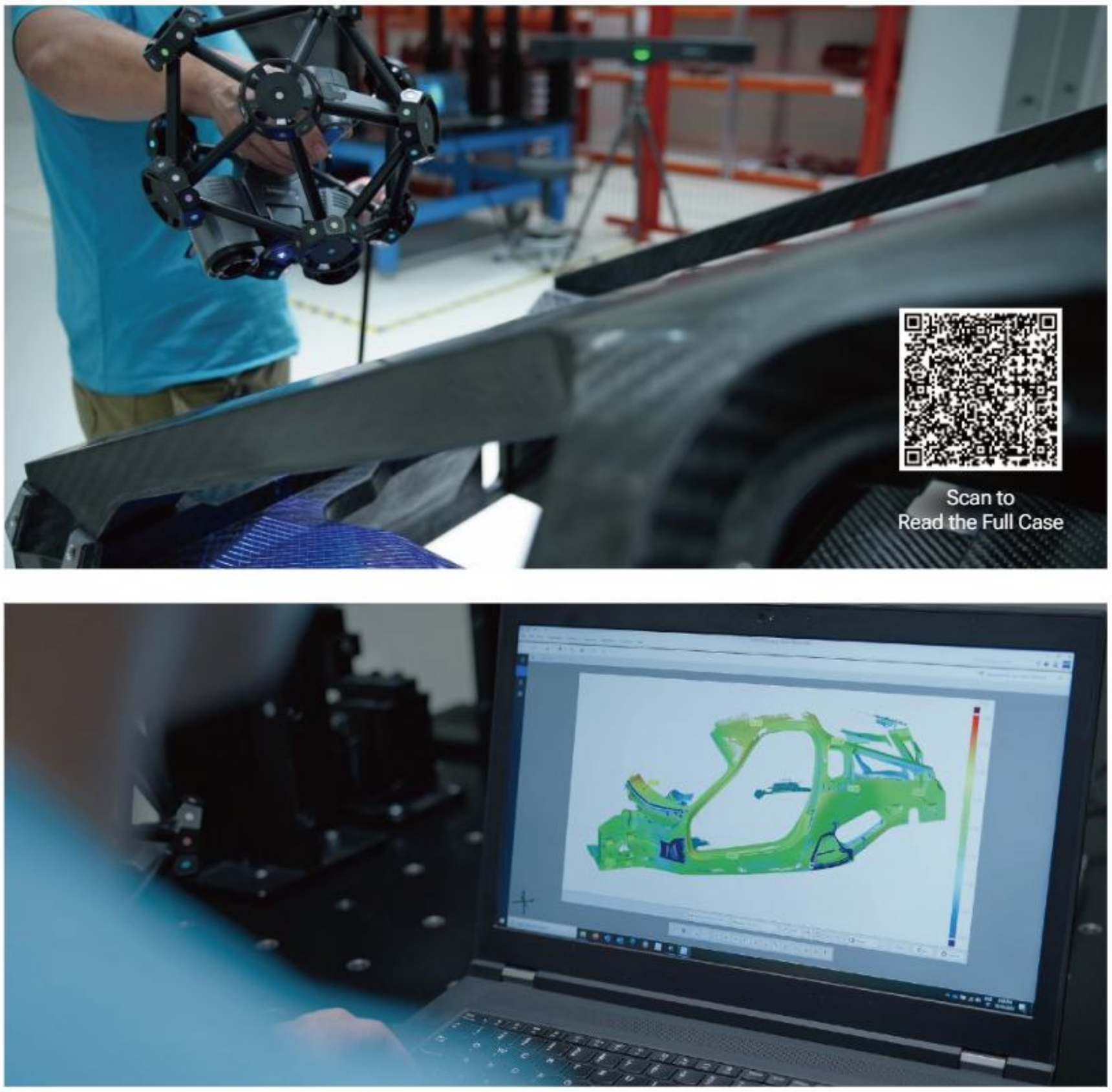
Automotive

In the realm of automotive modification, parts customization, and repairs, the prevailing challenges often manifest as time-consuming procedures, intricate complexities, and the arduous task of achieving absolute vehicle compatibility. Yet, through the utilization of cutting-edge technology such as the SHINING 3D scanner, a remarkable enhancement is achieved.

Application Fields

- Sheet Metal Inspection
- Prototype Design
- Molding Process Inspection
- Collision Testing and Safety Analysis
- Casting Component Check
- Parts Position and Fitting

Inspection of a Carbon Fiber Monocoque





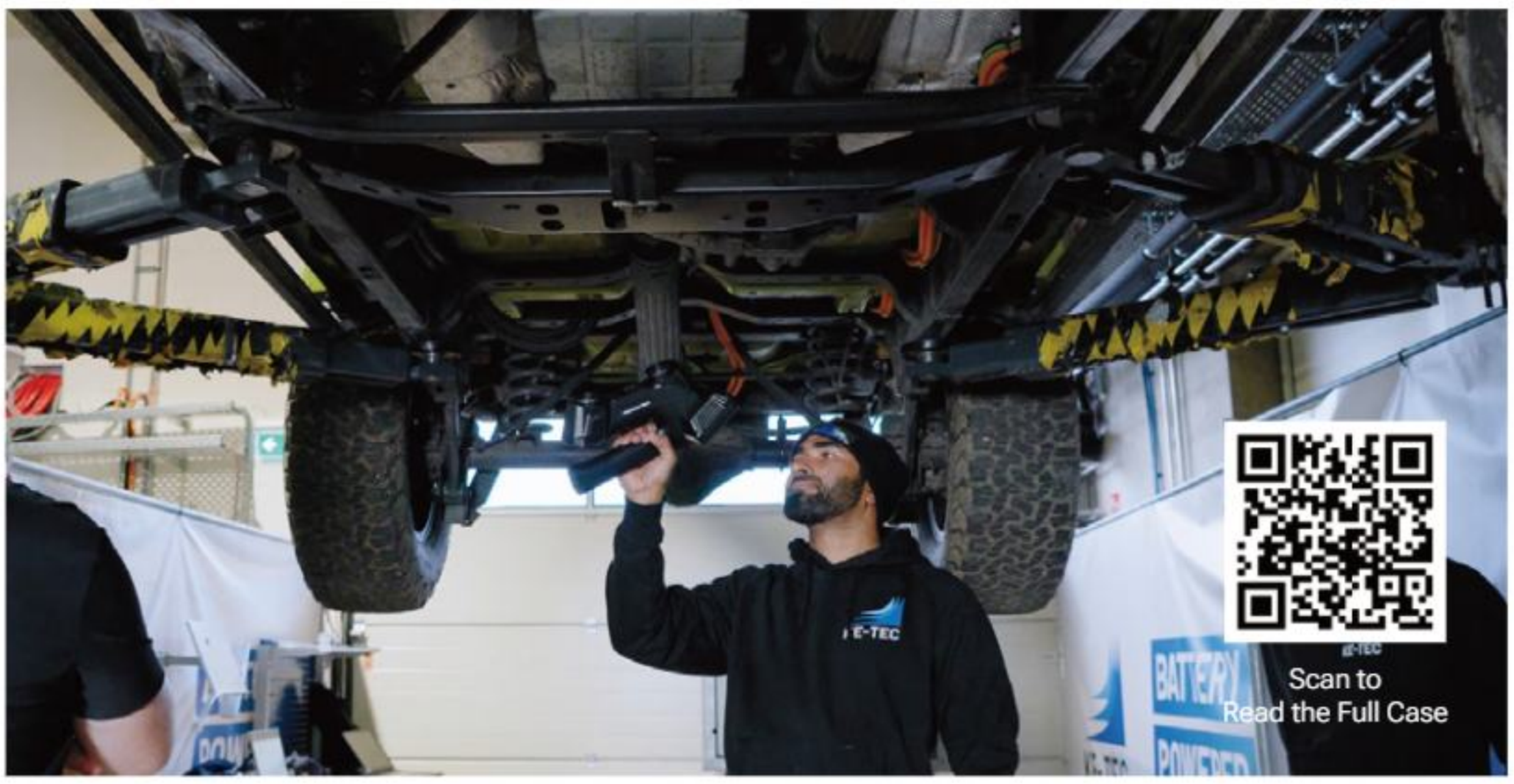
New Energy Vehicle

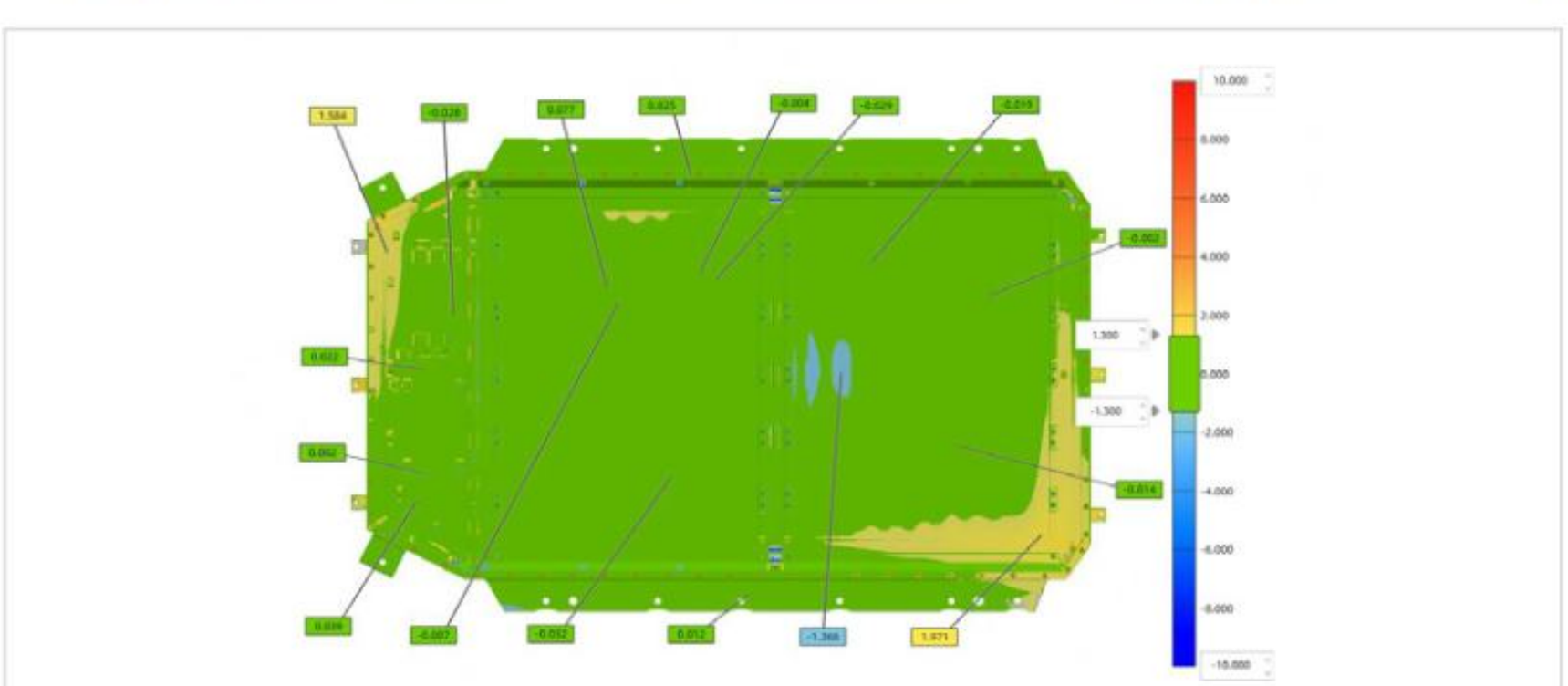
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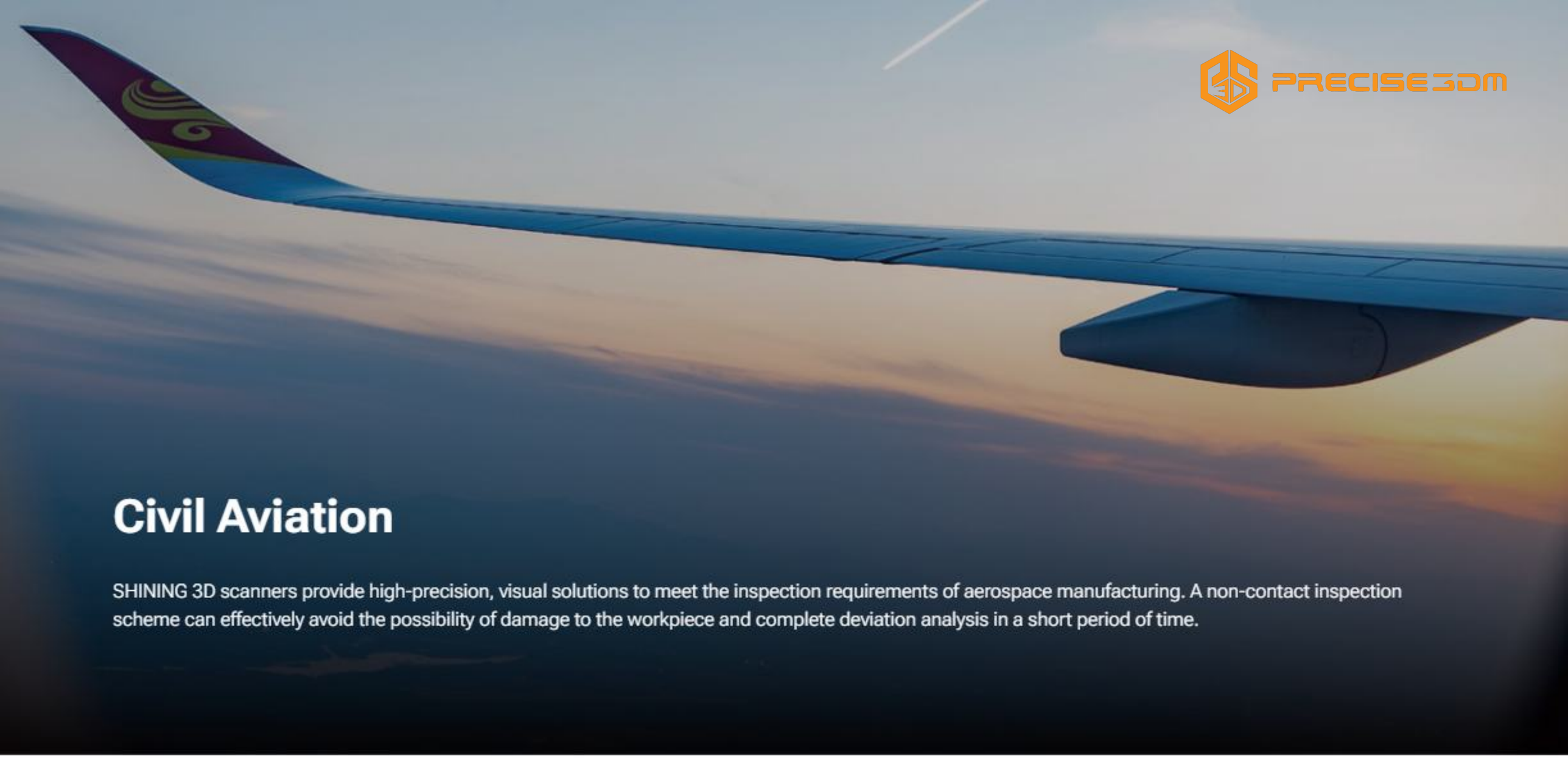
Application Fields

- Battery Pack Inspection
- Motor & Rotor Alignment
- ECU Measurement
- Cooling System Verification
- Structural Integration Analysis

On-Scanner 3D Inspection for EV Battery Development







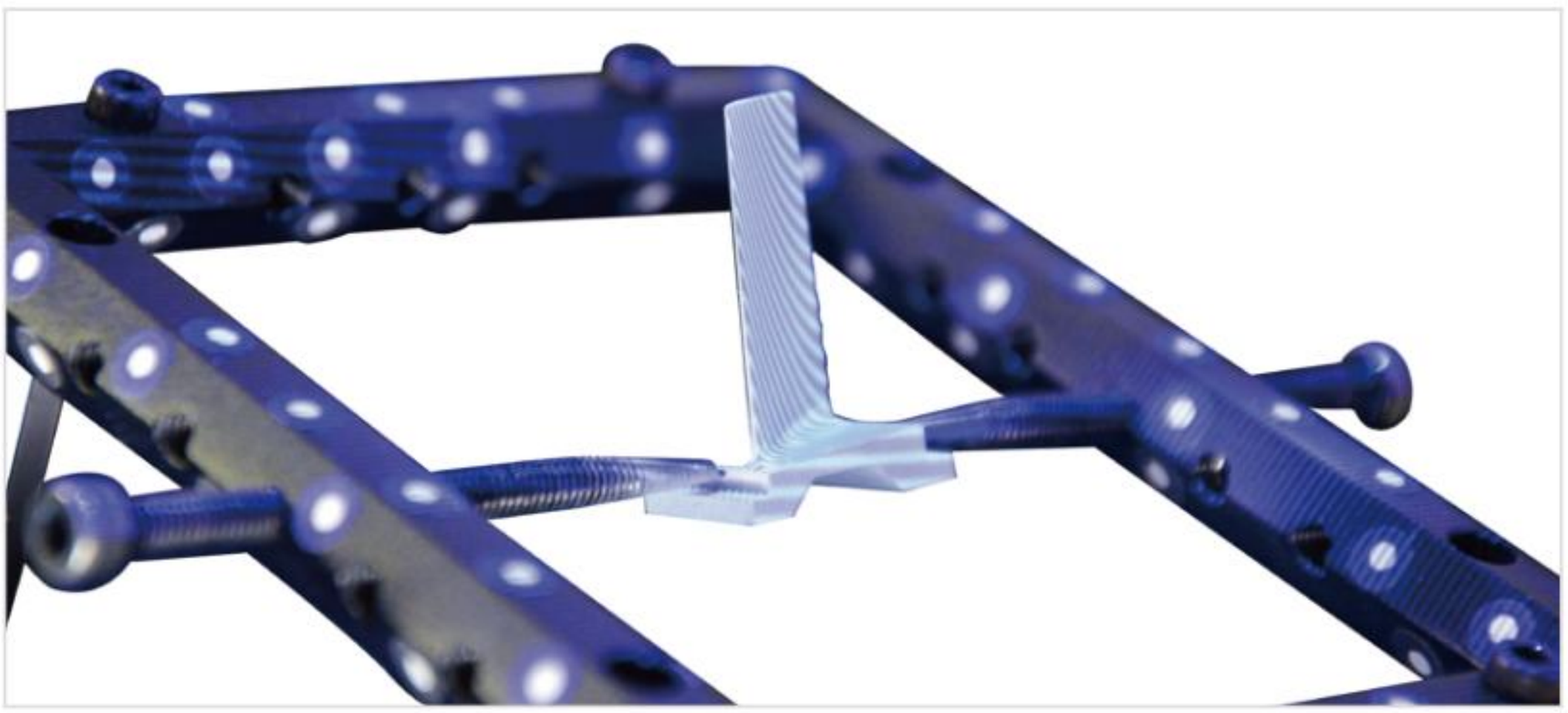
Civil Aviation

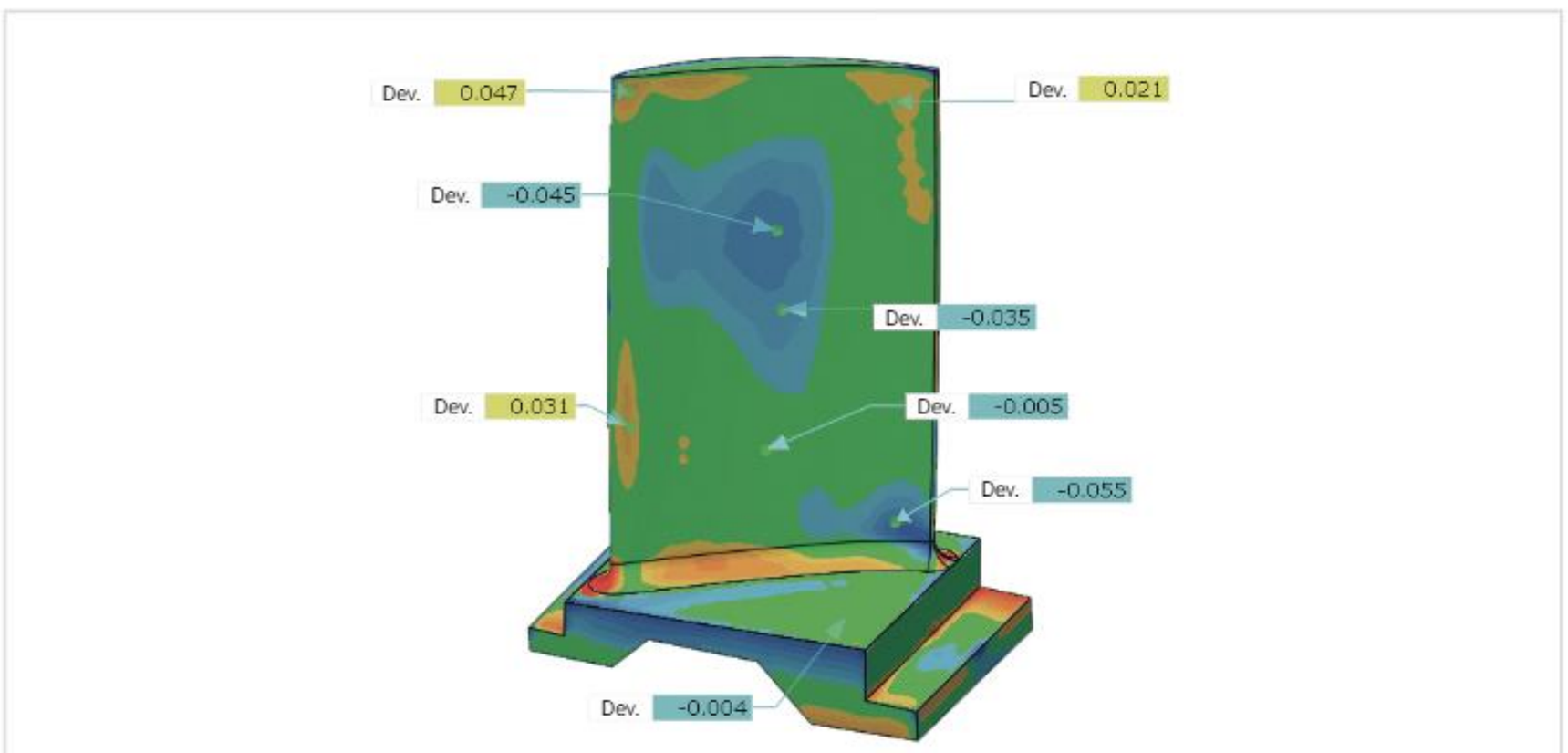
SHINING 3D scanners provide high-precision, visual solutions to meet the inspection requirements of aerospace manufacturing. A non-contact inspection scheme can effectively avoid the possibility of damage to the workpiece and complete deviation analysis in a short period of time.

Application Fields

- Component Digitization
- Tooling and Moulds Creation
- Dimensional Verification
- Blade Inspection
- Assembly Fit Analysis

Aero-engine Blade Inspection







Airplane MRO

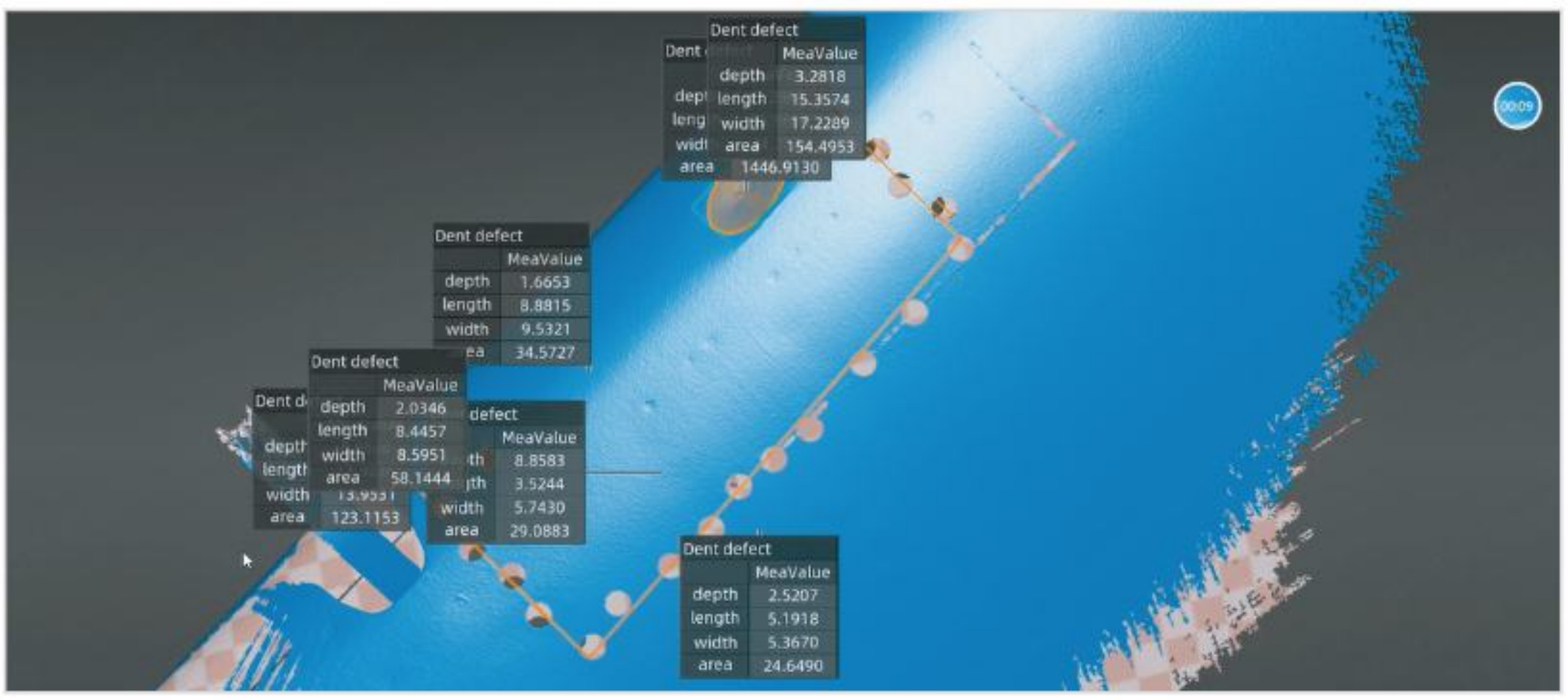
In aircraft maintenance, surface dents must be accurately detected and evaluated to ensure safety and aerodynamic performance. SHINING 3D solutions enable fast, precise 3D inspection and efficient repair verification.

Application Fields

- Dent Detection and Evaluation
- Engine Overhaul Analysis
- Surface Deformation Measurement
- Repair Verification
- Damage Reporting and Archiving

On-Ramp Wing Skin Damage Measurement







Marine

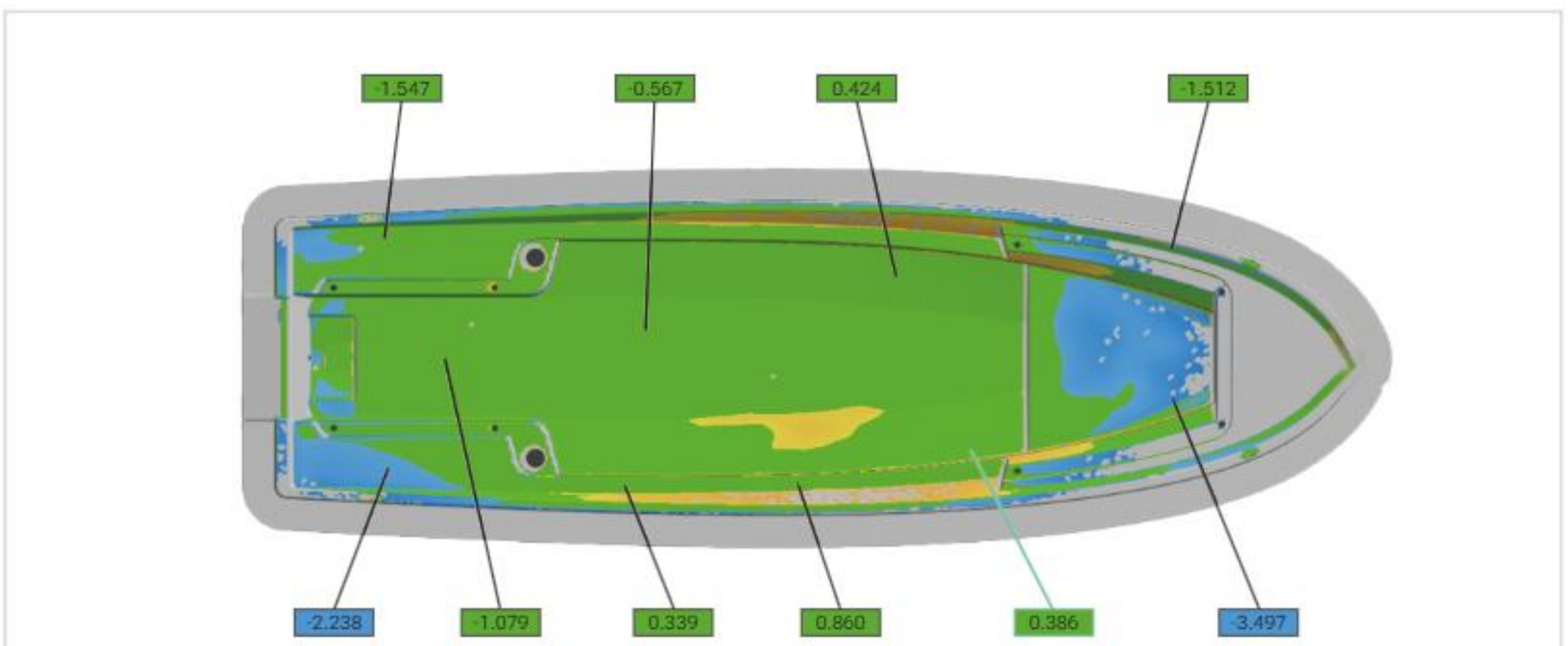
The marine sector demands high precision in the manufacturing, inspection, and maintenance of ships and offshore structures. Large structures, harsh environments, and complex surfaces present unique measurement challenges. SHINING 3D scanning solutions deliver fast, accurate, and non-contact data acquisition—ideal for hull deformation checks, structural assessments, and retrofit projects.

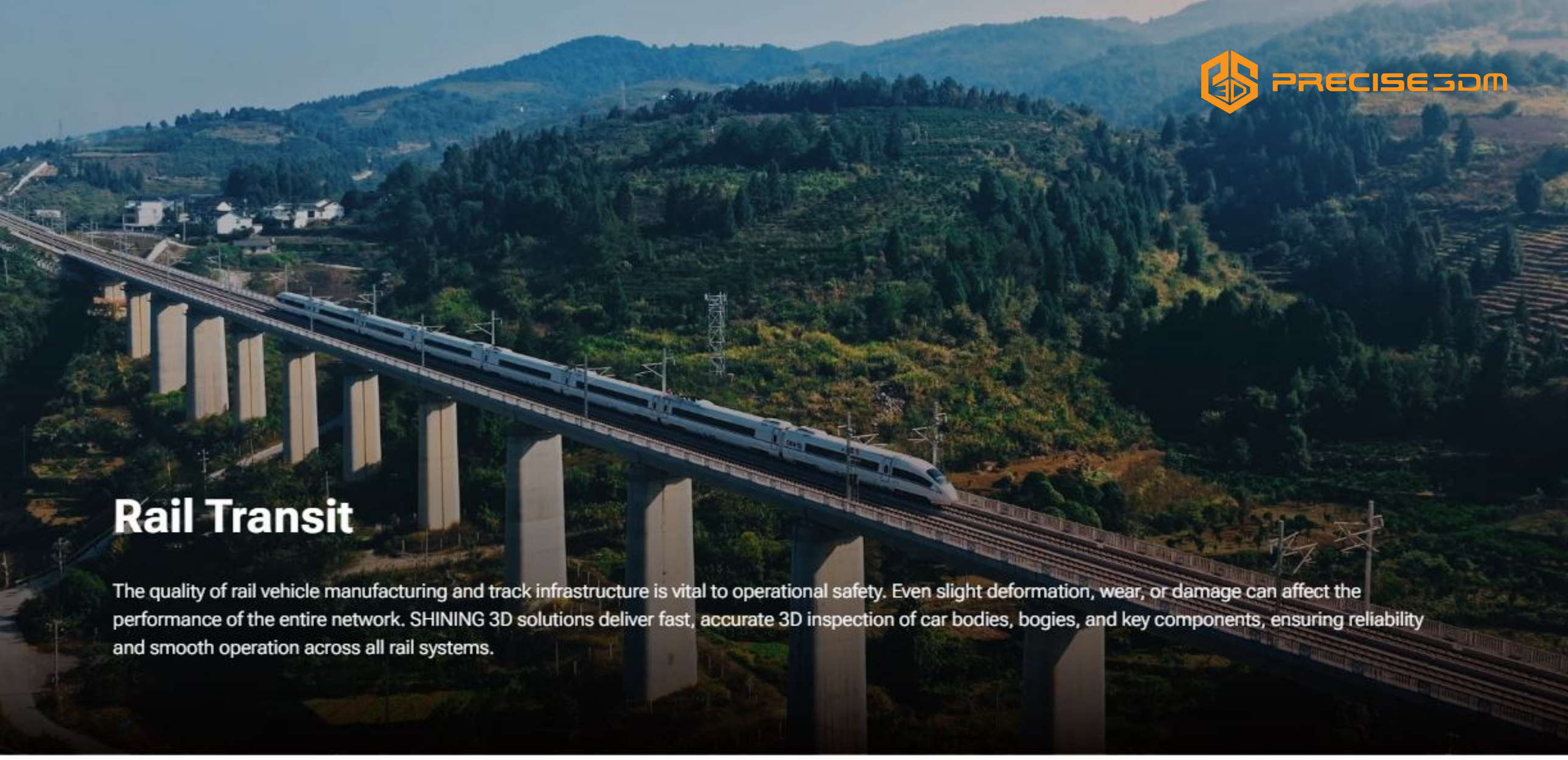
Application Fields

- Ship Hull Inspection and Deformation Analysis
- Propeller and Rudder Quality Control
- Retrofitting and Reverse Engineering
- Engine Room Layout and Piping Design
- Corrosion and Wear Monitoring
- Collision and Damage Assessment
- Customization and Modification

Marine Mold Inspection







Rail Transit

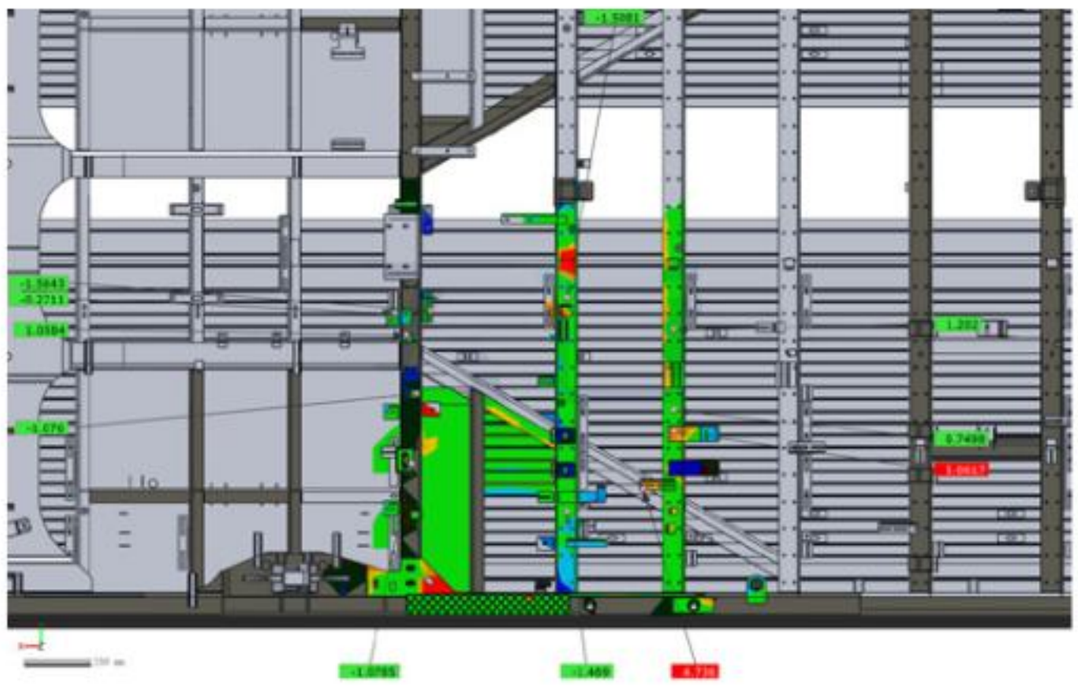
The quality of rail vehicle manufacturing and track infrastructure is vital to operational safety. Even slight deformation, wear, or damage can affect the performance of the entire network. SHINING 3D solutions deliver fast, accurate 3D inspection of car bodies, bogies, and key components, ensuring reliability and smooth operation across all rail systems.

Application Fields

- Cab and Nose Shape Inspection
- Window Assembly Deformation Analysis
- Body Weld Quality Inspection
- Axle Surface Scratch Inspection
- Coupler Casting Quality Check
- Turnout Dimensional Inspection

Body Weld Quality Inspection





Engineering Machinery

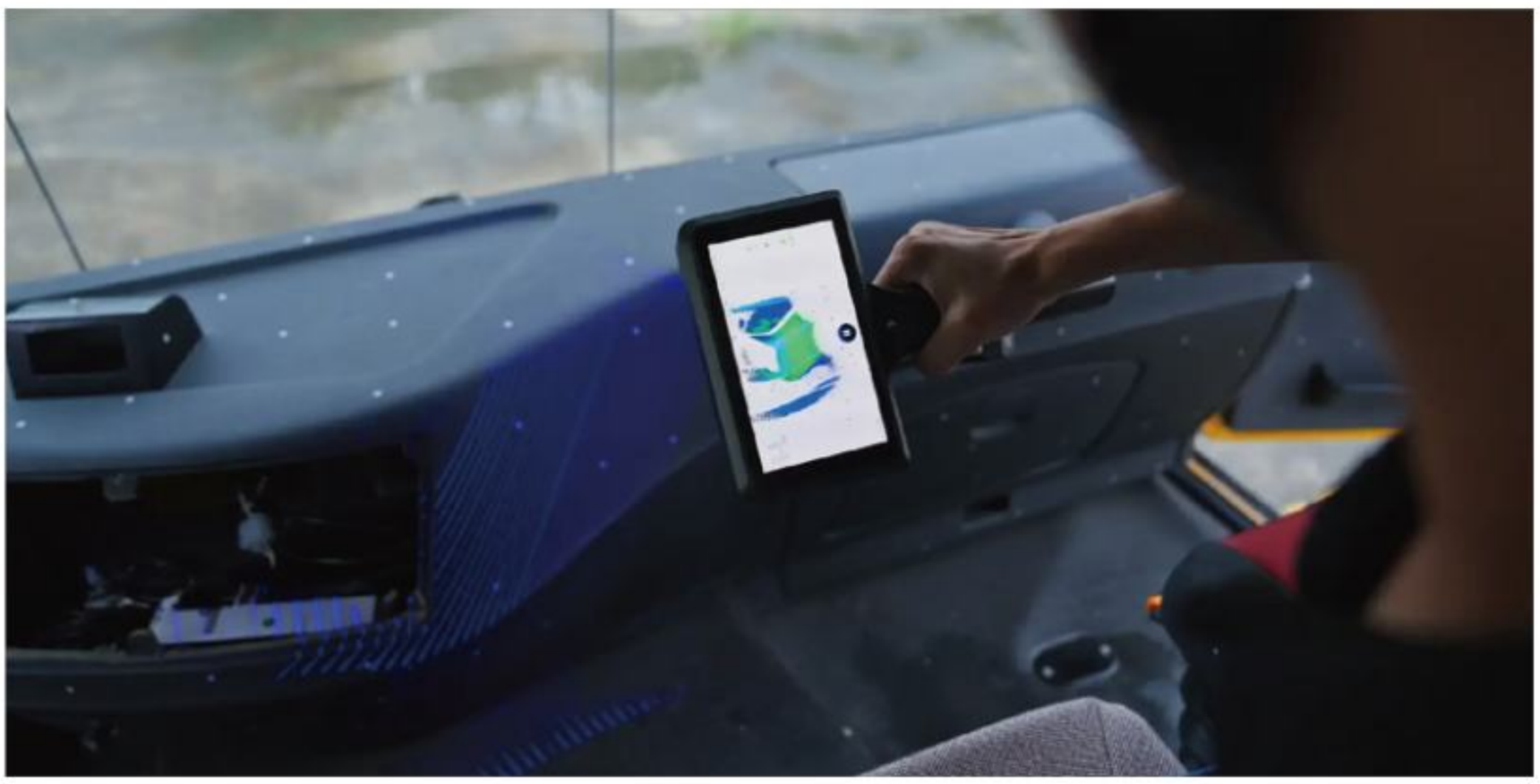
The engineering machinery industry involves the design, production, and maintenance of machinery used in construction and mining. Pain points include high maintenance costs, equipment downtime, and precision requirements. SHINING 3D scanners provide accurate measurements for wear and tear analysis, enabling predictive maintenance, and ensuring precise fitting of components, thereby reducing downtime and improving overall efficiency.

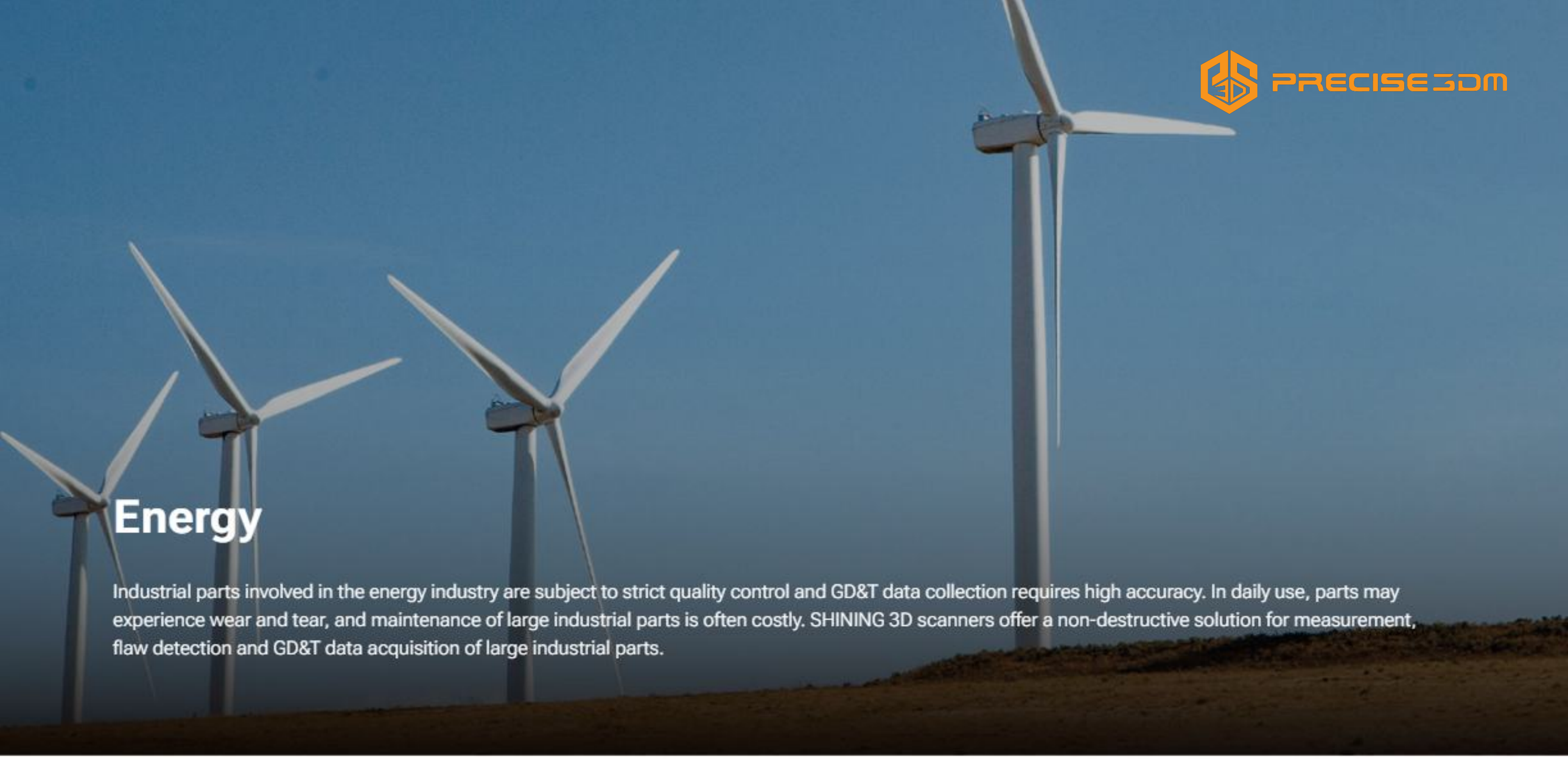
Application Fields

- Production Control
- Train Maintence
- Vibration Analysis
- Shipbuilding Quality Control
- Shape and Dimension Control

3D Scanning for Construction Machinery Manufacturing







Energy

Industrial parts involved in the energy industry are subject to strict quality control and GD&T data collection requires high accuracy. In daily use, parts may experience wear and tear, and maintenance of large industrial parts is often costly. SHINING 3D scanners offer a non-destructive solution for measurement, flaw detection and GD&T data acquisition of large industrial parts.

Application Fields

CFD Analysis

Maintenance Repare Overhaul

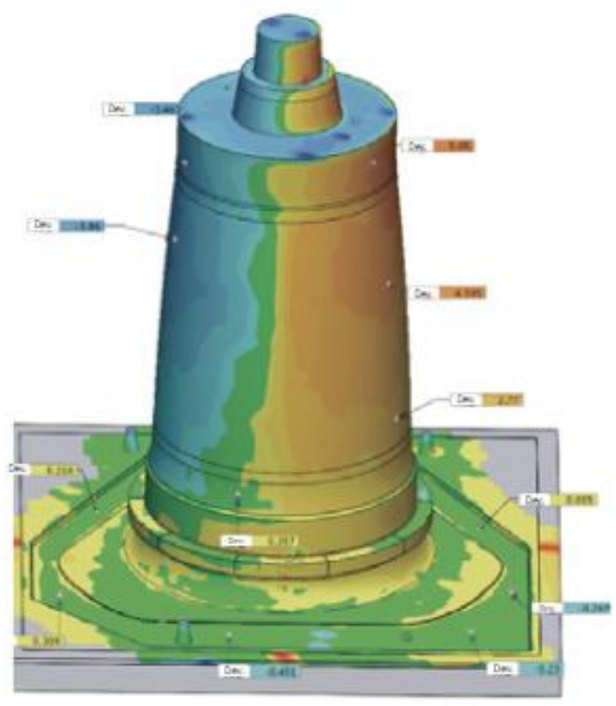
Gas & Turbines Analysis

Geometry and Construction Control

Water Turbines for Generation Inspect

Wind Power Equipment Mold Inspection





Electronic & Electrical

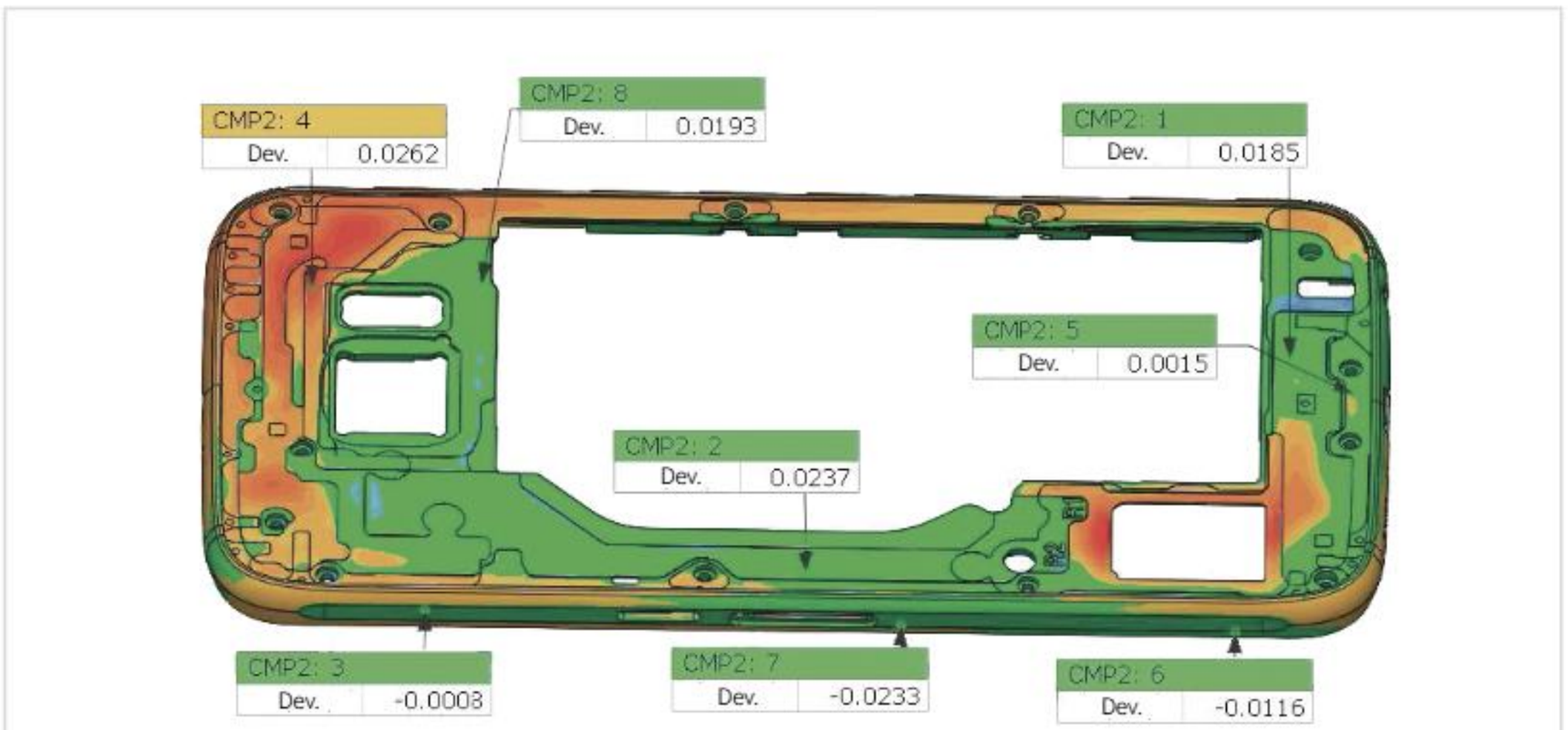
Quality control in consumer electronics production involves implementing measures to ensure that products meet specified quality standards throughout the manufacturing process. This typically includes various stages such as incoming materials inspection, in-process testing, and final product evaluation. SHINING 3D scanners are suitable for accurate 3D data acquisition of multi-category precision electronic parts, facilitating 3D full-size inspection.

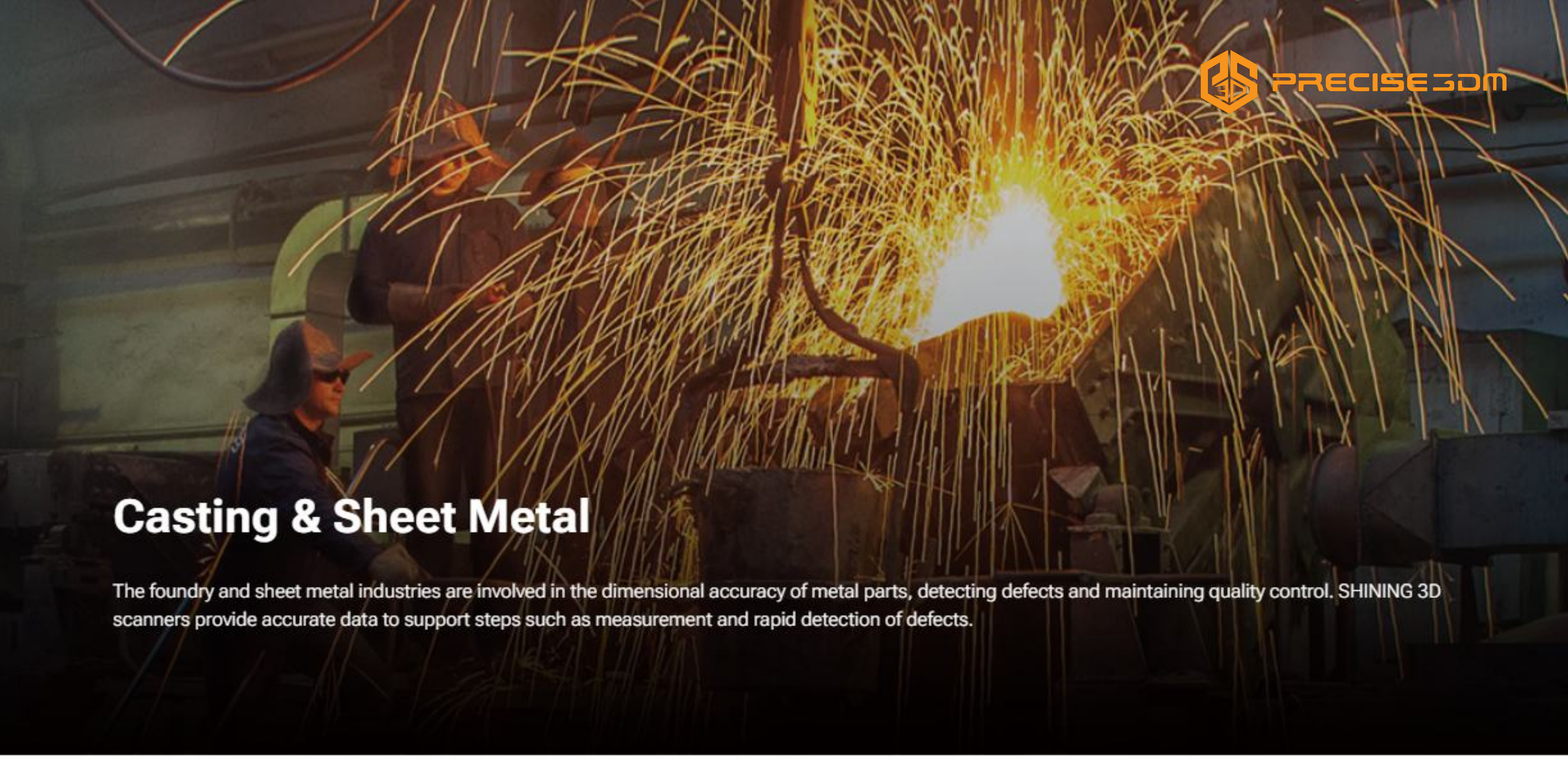
Application Fields

- PCB Inspection
- Component Placement Verification
- Thermal Analysis Using 3D Scanning
- Electronic Enclosures Modeling
- Electronic Components Reverse Engineering
- Consumer Electronics Virtual Prototyping

Smartphone Accessories Inspection







Casting & Sheet Metal

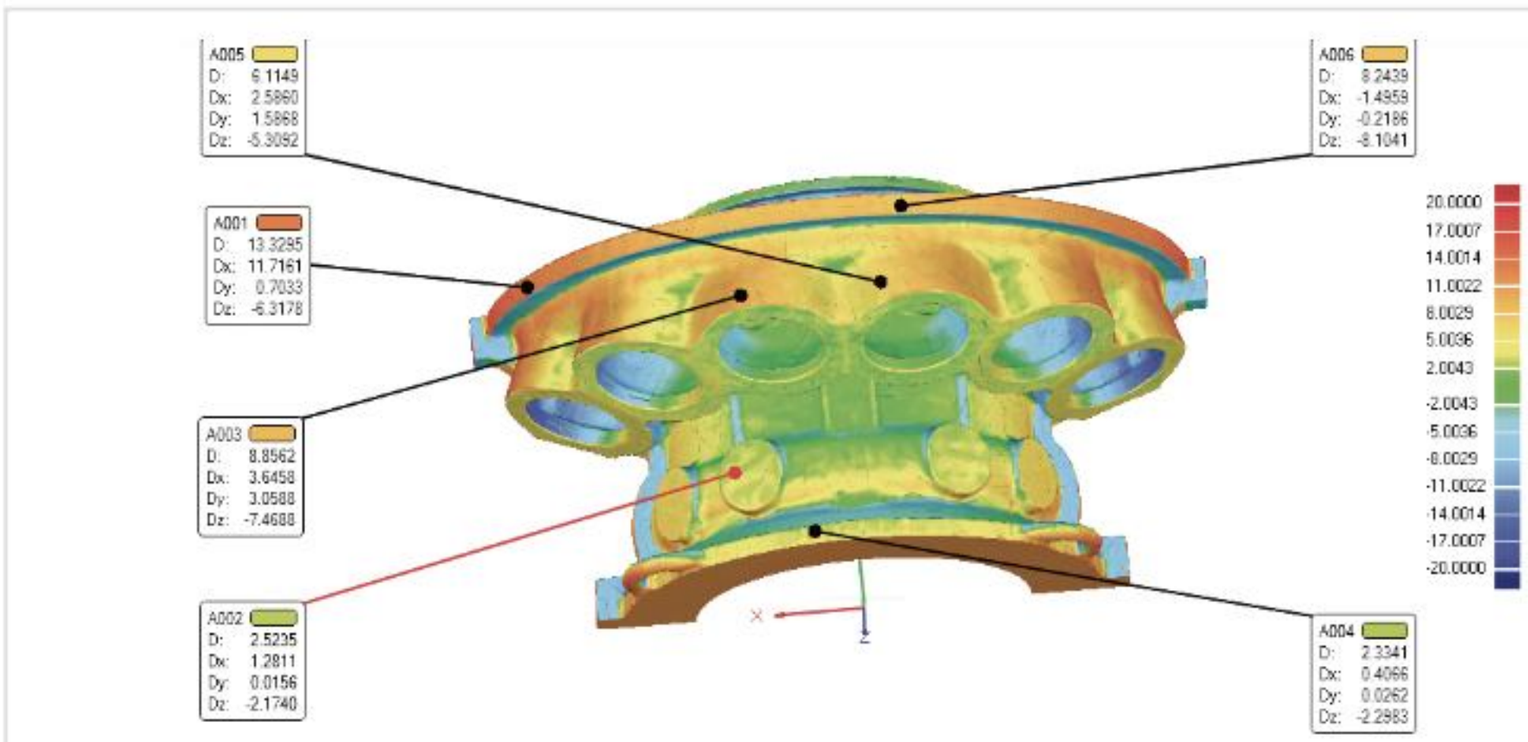
The foundry and sheet metal industries are involved in the dimensional accuracy of metal parts, detecting defects and maintaining quality control. SHINING 3D scanners provide accurate data to support steps such as measurement and rapid detection of defects.

Application Fields

- Parts Dimensional Inspection
- Surface Defect Detection
- Sheet Metal Forming Analysis
- Precision Casting Quality Control
- Metal Parts Reverse Engineering
- Tool Path Optimization Using 3D Models

Large Gas Turbine Castings Inspection





Tooling & Injection Molding

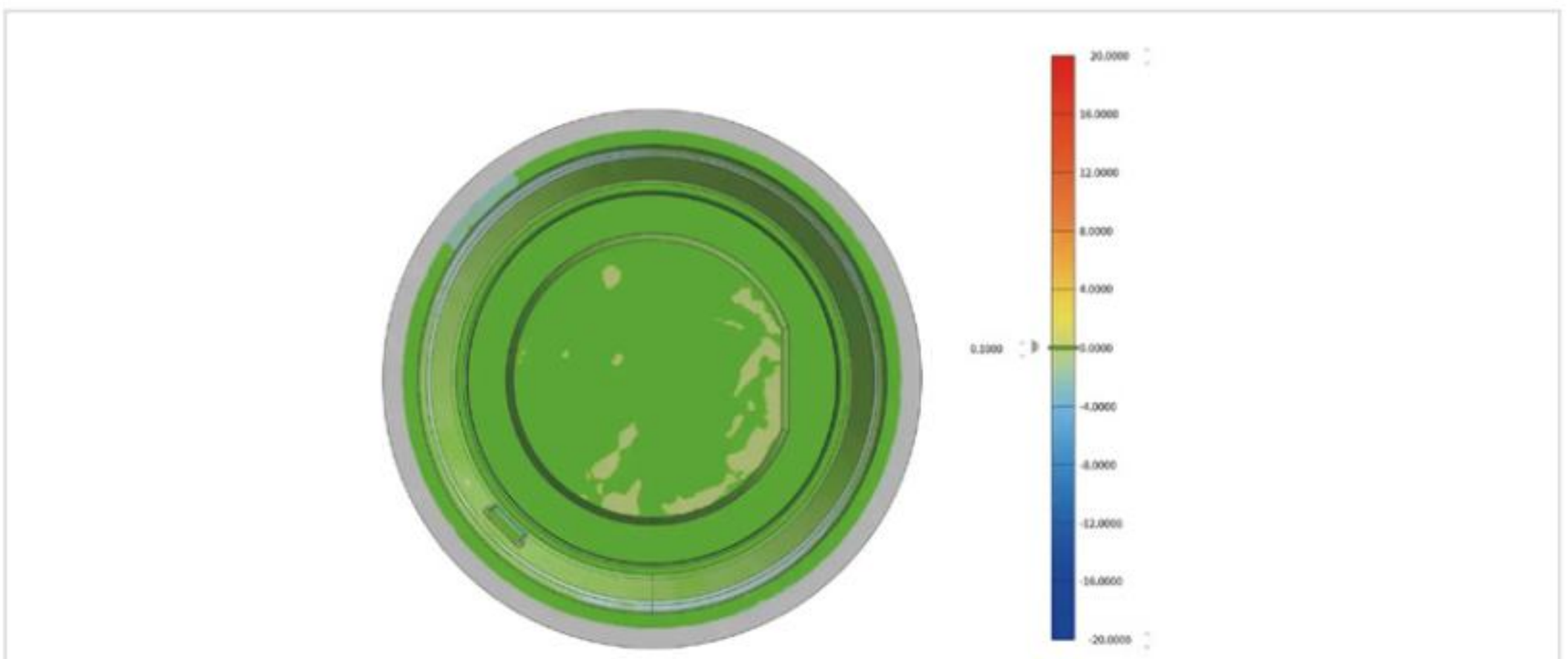
The Tooling & Injection Molding Industry involves creating molds and tools for producing plastic parts through injection molding. Key pain points include ensuring precision, reducing production time, and minimizing defects. SHINING 3D scanners provide accurate measurements for mold verification, enabling rapid prototyping, and ensuring quality control. This technology enhances precision, speeds up the development process, and reduces waste, ultimately improving overall efficiency and product quality.

Application Fields

- Mold & Tool Design Verification
- Part Validation
- Shrinkage and Warpage Analysis
- Mold Wear and Tear Monitoring
- First Article Inspection
- Virtual Simulation of Injection Molding Process

3D Scanning Improves Mold Inspection Efficiency





About SHINING 3D

Founded in 2004, SHINING 3D is committed to democratize the 3D scanning technology and provide the 3D digitizing solution based on the understanding of customer and market's demand, creating powerful products in ease of use and accessible.

Intellectual Property	Team Size	Scanners Sold
500+	1,300+	200,000+
SHINING 3D has independently researched and developed a number of core technologies in the 3D field.	SHINING 3D has **1300+** employees, **40%** of which are R&D staff.	SHINING 3D scanners are served to **55000+** users over **10000+** companies in more than **170** countries.

** Date till September, 2025*

Rigorous Product Quality Control System

ISO9001 ISO14001 ISO13485 MDSAP

CE FC UKCA KC FDA GMP

Valuing Data Security & Personal Information Protection

DNV DNV DNV DNV DJCP TISAX ENX ASSOCIATION

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